Centro de Excelencia de Redes y Seguridad

Conmutación y Enrutamiento (TI-203)

Producto 1

Proyecto de implementación empresarial

Red confiable, segura y escalable para la mediana empresa: VLANs, VTP, EtherChannel, STP, HSRP, OSPFv2 y ACLs sobre PNETLab

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Equipo 1: 172.16.0.0/15 | 2001:25:16::/48

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1. Resumen del proyecto, alcance y supuestos

El proyecto implementa una red para una mediana empresa con dos sitios: el **campus central** (R1-Core, dos switches multicapa SWM1/SWM2 en HSRP y tres switches de acceso) y una **sucursal** (R2 con switch de acceso SWA13 y una LAN de Datacenter). Todo corre en PNETLab con imágenes Cisco IOL. El documento sigue la estructura de los tres sprints del enunciado y cubre cada entregable.

1.1 Criterios de aceptación y dónde se cumplen

Criterio	Cómo se cumple	Pruebas
Conectividad entre VLANs según política	ACLs extendidas por VLAN y matriz de políticas (sección 4.4)	T01 – T14
Redundancia de gateway (HSRP) con failover probado	HSRPv2 IPv4/IPv6 en SWM1/SWM2, tracking del enlace al core y alineación con STP/OSPF	T15 – T20
Convergencia OSPF y propagación de ruta por defecto	OSPFv2 área única y multiárea (stub); R1-Core origina 0.0.0.0/0	T22 – T26
Accesos restringidos por ACLs	ACLs estándar y extendidas, VTY, NAT y filtro de borde	T07 – T14, T28, T29
Implementación en PNETLab	Guía de despliegue y tabla de cableado (secciones 2.2 y 4.8)	T30

1.2 Backlog (historias de usuario)

Historia	Descripción	Sección
HU1 – Direccionamiento	Plan IPv4 (VLSM) e IPv6 por departamento	2.4, 2.5, 2.6
HU2 – Segmentación VLAN	VLANs por departamento y mapa de puertos	2.7, 2.8, 3.2
HU3 – VTP	Dominios, modos servidor/cliente/transparente	2.7, 3.2

Historia	Descripción	Sección
HU4 – Redundancia L2/L3	EtherChannel, RPVST+, HSRP	3.2, 3.3, 3.4
HU5 – OSPFv2	Área única y multiárea	4.1, 4.2, 4.3
HU6 – Seguridad y ACLs	ACLs estándar y extendidas, NAT, VTY	4.4, 4.5
HU7 – Documentación y pruebas	Scripts de prueba, guía PNETLab, README	4.6 – 4.9

1.3 Supuestos y decisiones de diseño

- **SWM1/SWM2** son switches multicapa (IOL L3) y actúan como gateways HSRP de las VLANs del campus. R1-Core es el router de borde: sale a Internet, hace NAT/PAT y es ABR/ASBR de OSPF.
- **Datacenter** y **P2P** se tratan como LANs (no VLANs), tal como indica el enunciado: la LAN Datacenter cuelga de R2 e0/1 y el P2P es el enlace serial R1-Core ↔ R2.
- **Blackhull** (VLAN 999) es la VLAN "agujero negro": VLAN nativa de los trunks y destino de los puertos sin uso (apagados). Tiene subred reservada pero **sin gateway** ni SVI. Se define en ambos dominios VTP.
- **Tamaño de subredes:** hosts + direcciones de gateway (1 por router; 3 por VLAN con HSRP: VIP + SWM1 + SWM2) + red + broadcast. Por eso Gestión (29 hosts + 3 HSRP) usa /26 en vez de /27.
- **IPv6:** un /64 por red, asignado en el mismo orden de tamaño que el plan IPv4 (la red de 1500 hosts es la primera: 2001:25:16::/64).
- **ISP:** se simula con un enlace 203.0.113.0/30 (rango de documentación). Si en PNETLab usas un Cloud con DHCP/NAT, cambia la IP de e0/2 por `ip address dhcp` y la ruta por defecto (ver 4.5).
- **Contraseñas** de laboratorio: usuario `admin` y clave `Itla2026` (enable, VTP, HSRP, OSPF
- **Nombres de interfaz** tipo IOL (Ethernet0/0, Serial1/0). Si tus imágenes tienen menos puertos, ajusta los rangos de "puertos sin uso".

2. Sprint 1 – Diseño y plan básico

2.1 Lista de nodos

Nodo	Tipo	Plantilla PNETLab	Rol	Puertos usados
ISP	Nube	Network > Cloud / NAT	Salida a Internet	e0/0
R1-Core	Router	Cisco IOL (L3)	Core, ABR/ASBR, NAT, DHCP	e0/0, e0/1, e0/2, s1/0
R2	Router	Cisco IOL (L3)	Sucursal: gateway FIN/ACA/DC, DHCP	e0/0 (subif), e0/1, s1/0
SWM1	Switch L3	Cisco IOL L3	Gateway HSRP, VTP server, root STP 10/30	e0/0, e0/1-3, e1/0-2
SWM2	Switch L3	Cisco IOL L3	Gateway HSRP, VTP server, root STP 20/40	e0/0, e0/1-3, e1/0-2
SWAC1	Switch L2	Cisco IOL L2	Acceso campus (Docentes)	e0/0, e0/1, e1/0
SWAC2	Switch L2	Cisco IOL L2	Acceso campus (Estudiantes)	e0/0, e0/1, e2/0
SWAC3	Switch L2	Cisco IOL L2	Acceso campus (Gestión)	e0/0, e0/1, e3/0
SWA13	Switch L2	Cisco IOL L2	Acceso sucursal (VLAN 50/60)	e0/0, e1/0, e2/0
SW	Switch L2	Cisco IOL L2	Switch del Datacenter	e0/0, e0/1
DOC1, EST1, GES1, ACA1, FIN1	PC	VPCS	Equipos de prueba (DHCP)	eth0
SERVICIOS	Servidor	VPCS o Linux	Servidor del Datacenter (172.16.10.10)	e0 / eth0

2.2 Lista de conexiones (cableado en PNETLab)

#	Extremo A	Puerto A	Extremo B	Puerto B	Función
1	SERVICIOS	e0	SW	e0/1	Acceso VLAN 70 (Datacenter)
2	SW	e0/0	R2	e0/1	LAN Datacenter 172.16.10.0/26
3	R2	e0/0	SWA13	e0/0	Trunk 802.1Q VLAN 50 y 60 (router-on-a-stick)
4	SWA13	e1/0	ACA1	eth0	Acceso VLAN 60

#	Extremo A	Puerto A	Extremo B	Puerto B	Función
5	SWA13	e2/0	FIN1	eth0	Acceso VLAN 50
6	R2	s1/0	R1-Core	s1/0	P2P serial 172.16.10.192/30
7	R1-Core	e0/2	ISP	e0/0	Salida a Internet 203.0.113.0/30
8	R1-Core	e0/0	SWM1	e0/0	Enlace L3 172.16.10.196/30
9	R1-Core	e0/1	SWM2	e0/0	Enlace L3 172.16.10.200/30
10	SWM1	e0/1-3	SWM2	e0/1-3	EtherChannel Po1 (LACP), trunk
11	SWM1	e1/0	SWAC1	e0/0	Trunk
12	SWM1	e1/1	SWAC2	e0/0	Trunk
13	SWM1	e1/2	SWAC3	e0/0	Trunk
14	SWM2	e1/0	SWAC1	e0/1	Trunk
15	SWM2	e1/1	SWAC2	e0/1	Trunk
16	SWM2	e1/2	SWAC3	e0/1	Trunk
17	SWAC1	e1/0	DOC1	eth0	Acceso VLAN 20 (+ voz 40)
18	SWAC2	e2/0	EST1	eth0	Acceso VLAN 10 (+ voz 40)
19	SWAC3	e3/0	GES1	eth0	Acceso VLAN 30 (+ voz 40)

2.3 Diagramas de topología

Figura 1. Topología física con nombres de interfaz (equivale a lo que se arma en PNETLab).

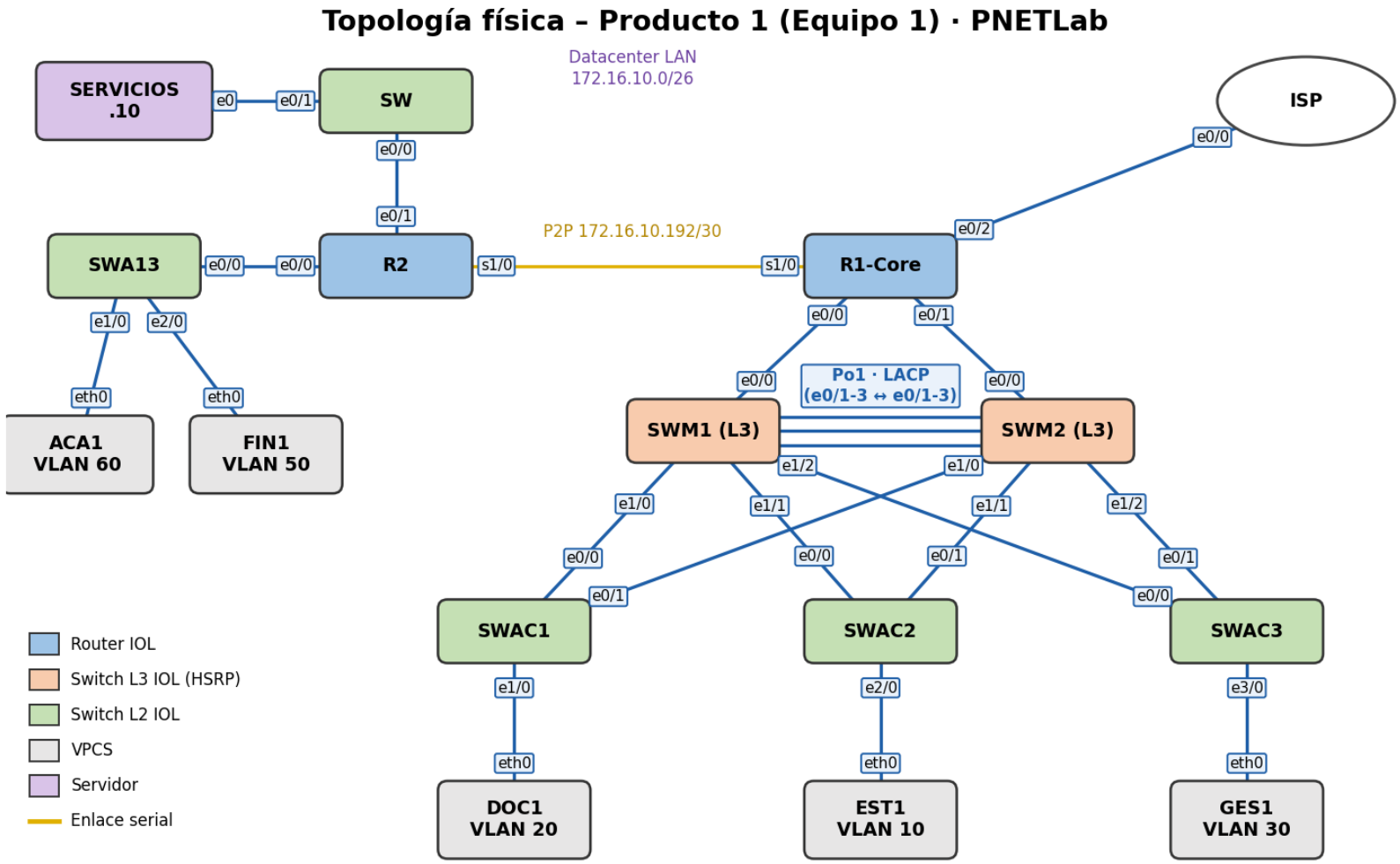
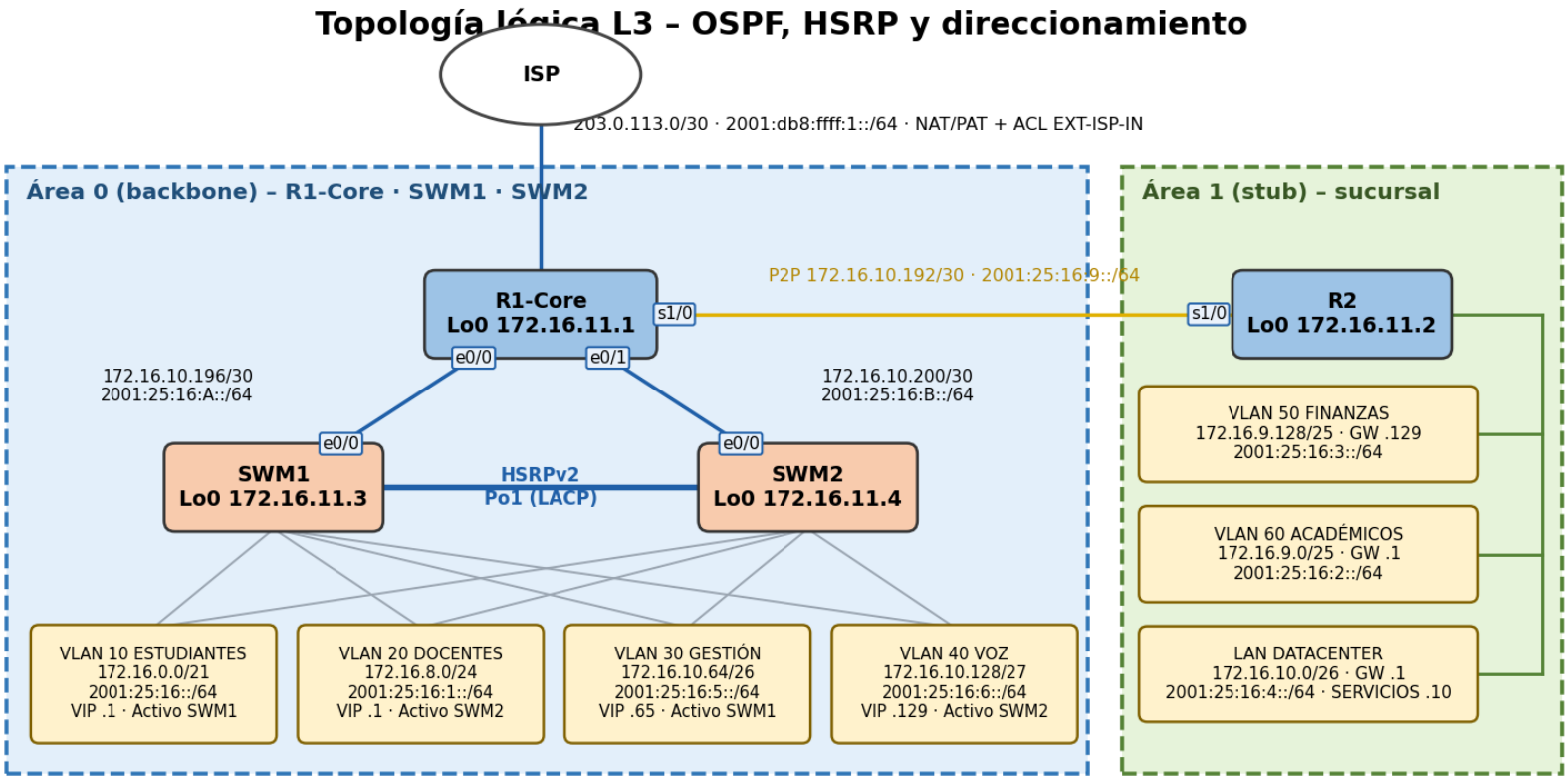


Figura 2. Topología lógica L3: áreas OSPF, grupos HSRP, subredes IPv4/IPv6 y loopbacks.



Rutas: R1-Core origina 0.0.0.0/0 (estática hacia ISP) y la propaga por OSPF · IDs OSPF = Loopback0 · costo SVI 10 (activo) / 20 (standby) para alinear OSPF con HSRP

Blackhull R1-Core: 172.16.10.160/28 · 2001:25:16:7::/64 (VLAN 999, sin gateway) | Blackhull R2: 172.16.10.176/28 · 2001:25:16:8::/64 (VLAN 999, sin gateway)

Reservas: 172.16.10.204/30 (enlaces futuros) · 172.16.11.0/24 (loopbacks) · 172.16.12.0 – 172.17.255.255 (crecimiento multiárea)

2.4 Plan de direccionamiento IPv4 (VLSM) – bloque 172.16.0.0/15

El bloque del equipo cubre 172.16.0.0 – 172.17.255.255. Se ordenan las redes de mayor a menor y se asigna cada subred en el primer límite de bloque libre. Dirección necesaria = hosts + gateways + 2 (red y broadcast).

Cálculo del tamaño

Red	Hosts	Dir. necesarias	Bits host	Prefijo	Máscara	Hosts útiles
Estudiantes	1500	1500+3+2 = 1505	11	/21	255.255.248.0	2046
Docentes	164	164+3+2 = 169	8	/24	255.255.255.0	254
Académicos (R2)	80	80+1+2 = 83	7	/25	255.255.255.128	126
Finanzas (R2)	64	64+1+2 = 67	7	/25	255.255.255.128	126
Datacenter (R2)	44	44+1+2 = 47	6	/26	255.255.255.192	62
Gestión	29	29+3+2 = 34	6	/26	255.255.255.192	62
Voz	16	16+3+2 = 21	5	/27	255.255.255.224	30
Blackhull R1-Core	10	10+2 = 12	4	/28	255.255.255.240	14
Blackhull R2	10	10+2 = 12	4	/28	255.255.255.240	14
P2P R1-Core ↔ R2	2	2+2 = 4	2	/30	255.255.255.252	2

Asignación de subredes

Orden	Red	Subred	Rango útil	Broadcast	Gateway
1	Estudiantes	172.16.0.0/21	172.16.0.1 – 172.16.7.254	172.16.7.255	VIP .0.1
2	Docentes	172.16.8.0/24	172.16.8.1 – 172.16.8.254	172.16.8.255	VIP .8.1
3	Académicos	172.16.9.0/25	172.16.9.1 – 172.16.9.126	172.16.9.127	172.16.9.1
4	Finanzas	172.16.9.128/25	172.16.9.129 – 172.16.9.254	172.16.9.255	172.16.9.129
5	Datacenter	172.16.10.0/26	172.16.10.1 – 172.16.10.62	172.16.10.63	172.16.10.1

Orden	Red	Subred	Rango útil	Broadcast	Gateway
6	Gestión	172.16.10.64/26	172.16.10.65 – 172.16.10.126	172.16.10.127	VIP .10.65
7	Voz	172.16.10.128/27	172.16.10.129 – 172.16.10.158	172.16.10.159	VIP .10.129
8	Blackhull R1-Core	172.16.10.160/28	172.16.10.161 – 172.16.10.174	172.16.10.175	Sin gateway
9	Blackhull R2	172.16.10.176/28	172.16.10.177 – 172.16.10.190	172.16.10.191	Sin gateway
10	P2P R1-Core ↔ R2	172.16.10.192/30	172.16.10.193 – 172.16.10.194	172.16.10.195	—

Enlaces de infraestructura, loopbacks y reservas

Enlace	Subred	Extremo A	Extremo B	Nota
R1-Core ↔ SWM1	172.16.10.196/30	R1-Core .197	SWM1 .198	Ruteado, OSPF área 0
R1-Core ↔ SWM2	172.16.10.200/30	R1-Core .201	SWM2 .202	Ruteado, OSPF área 0
Reserva enlaces	172.16.10.204/30	—	—	Crecimiento
R1-Core ↔ ISP	203.0.113.0/30	ISP .1	R1-Core .2	Rango de documentación
Loopbacks /32	172.16.11.0/24	R1-Core .1, R2 .2	SWM1 .3, SWM2 .4	ID de OSPF y gestión
Crecimiento multiárea	172.16.12.0 – 172.17.255.255	—	—	Sin asignar

2.5 Plan de direccionamiento IPv6 – 2001:25:16::/48

Cada red recibe un /64 en el mismo orden de tamaño del plan IPv4. Las redes de infraestructura usan los identificadores de subred A, B y FF.

Orden	Red	Prefijo IPv6	IPv4 equivalente	Método de asignación
1	Estudiantes	2001:25:16::/64	172.16.0.0/21	SLAAC + DHCPv6 sin estado
2	Docentes	2001:25:16:1::/64	172.16.8.0/24	SLAAC + DHCPv6 sin estado
3	Académicos	2001:25:16:2::/64	172.16.9.0/25	SLAAC + DHCPv6 sin estado (R2)
4	Finanzas	2001:25:16:3::/64	172.16.9.128/25	DHCPv6 con estado (R2, flag M)

Orden	Red	Prefijo IPv6	IPv4 equivalente	Método de asignación
5	Datacenter	2001:25:16:4::/64	172.16.10.0/26	Estático (SERVICIOS ::10)
6	Gestión	2001:25:16:5::/64	172.16.10.64/26	SLAAC + DHCPv6 sin estado
7	Voz	2001:25:16:6::/64	172.16.10.128/27	SLAAC + DHCPv6 sin estado
8	Blackhull R1-Core	2001:25:16:7::/64	172.16.10.160/28	Sin uso
9	Blackhull R2	2001:25:16:8::/64	172.16.10.176/28	Sin uso
10	P2P R1-Core ↔ R2	2001:25:16:9::/64	172.16.10.192/30	Estático ::1 / ::2
11	R1-Core ↔ SWM1	2001:25:16:A::/64	172.16.10.196/30	Estático ::1 / ::2
12	R1-Core ↔ SWM2	2001:25:16:B::/64	172.16.10.200/30	Estático ::1 / ::2
13	Loopbacks	2001:25:16:FF::/64	172.16.11.0/24	/128: R1-Core ::1, R2 ::2, SWM1 ::3, SWM2 ::4

Plan SLAAC/DHCPv6: los routers anuncian el prefijo por RA (SLAAC) y activan el flag O (`ipv6 nd other-config-flag`) para que los hosts pidan DNS y dominio por DHCPv6 sin estado. En Finanzas se usa DHCPv6 con estado (flag M, prefijo anunciado sin autoconfiguración) para llevar control de asignaciones. En las VLANs con HSRP la IP de gateway que ven los hosts es la dirección link-local virtual (`standby … ipv6 autoconfig`).

2.6 Tabla de direccionamiento por dispositivo

Dispositivo	Interfaz	IPv4 / máscara	IPv6 / prefijo	Gateway / nota
R1-Core	e0/0	172.16.10.197/30	2001:25:16:A::1/64	a SWM1
R1-Core	e0/1	172.16.10.201/30	2001:25:16:B::1/64	a SWM2
R1-Core	e0/2	203.0.113.2/30	2001:db8:ffff:1::2/64	GW 203.0.113.1
R1-Core	s1/0	172.16.10.193/30	2001:25:16:9::1/64	a R2
R1-Core	Lo0	172.16.11.1/32	2001:25:16:FF::1/128	Router-ID
R2	e0/0.50	172.16.9.129/25	2001:25:16:3::1/64	GW Finanzas
R2	e0/0.60	172.16.9.1/25	2001:25:16:2::1/64	GW Académicos
R2	e0/1	172.16.10.1/26	2001:25:16:4::1/64	GW Datacenter
R2	s1/0	172.16.10.194/30	2001:25:16:9::2/64	a R1-Core
R2	Lo0	172.16.11.2/32	2001:25:16:FF::2/128	Router-ID

Dispositivo	Interfaz	IPv4 / máscara	IPv6 / prefijo	Gateway / nota
SWM1	e0/0	172.16.10.198/30	2001:25:16:A::2/64	a R1-Core
SWM1	Vlan10	172.16.0.2/21	2001:25:16::2/64	VIP 172.16.0.1
SWM1	Vlan20	172.16.8.2/24	2001:25:16:1::2/64	VIP 172.16.8.1
SWM1	Vlan30	172.16.10.66/26	2001:25:16:5::2/64	VIP 172.16.10.65
SWM1	Vlan40	172.16.10.130/27	2001:25:16:6::2/64	VIP 172.16.10.129
SWM1	Lo0	172.16.11.3/32	2001:25:16:FF::3/128	Router-ID
SWM2	e0/0	172.16.10.202/30	2001:25:16:B::2/64	a R1-Core
SWM2	Vlan10	172.16.0.3/21	2001:25:16::3/64	VIP 172.16.0.1
SWM2	Vlan20	172.16.8.3/24	2001:25:16:1::3/64	VIP 172.16.8.1
SWM2	Vlan30	172.16.10.67/26	2001:25:16:5::3/64	VIP 172.16.10.65
SWM2	Vlan40	172.16.10.131/27	2001:25:16:6::3/64	VIP 172.16.10.129
SWM2	Lo0	172.16.11.4/32	2001:25:16:FF::4/128	Router-ID
SWAC1	Vlan30	172.16.10.68/26	—	GW 172.16.10.65
SWAC2	Vlan30	172.16.10.69/26	—	GW 172.16.10.65
SWAC3	Vlan30	172.16.10.70/26	—	GW 172.16.10.65
SWA13	Vlan60	172.16.9.126/25	—	GW 172.16.9.1
SW	Vlan70	172.16.10.2/26	—	GW 172.16.10.1
SERVICIOS	eth	172.16.10.10/26	2001:25:16:4::10/64	GW 172.16.10.1
DOC1, EST1, GES1, ACA1, FIN1	eth0	DHCP (pool de su VLAN)	SLAAC (FIN1: DHCPv6)	GW = VIP / gateway

2.7 Tabla de VLANs y dominios VTP

ID	Nombre	Dominio VTP	Subred	Gateway	Dónde se enruta
10	ESTUDIANTES	ITLA-CORE	172.16.0.0/21	172.16.0.1	SVI en SWM1/SWM2 (HSRP)
20	DOCENTES	ITLA-CORE	172.16.8.0/24	172.16.8.1	SVI en SWM1/SWM2 (HSRP)

ID	Nombre	Dominio VTP	Subred	Gateway	Dónde se enruta
30	GESTION	ITLA-CORE	172.16.10.64/26	172.16.10.65	SVI en SWM1/SWM2 (HSRP)
40	VOZ	ITLA-CORE	172.16.10.128/27	172.16.10.129	SVI en SWM1/SWM2 (HSRP)
999	BLACKHULL	ITLA-CORE	172.16.10.160/28	—	Sin SVI; nativa y puertos sin uso
50	FINANZAS	ITLA-R2	172.16.9.128/25	172.16.9.129	R2 e0/0.50
60	ACADEMICOS	ITLA-R2	172.16.9.0/25	172.16.9.1	R2 e0/0.60
999	BLACKHULL	ITLA-R2	172.16.10.176/28	—	Sin subinterfaz; nativa y puertos sin uso
70	DATACENTER	Local en SW	172.16.10.0/26	172.16.10.1	LAN Datacenter: R2 e0/1 (sin trunk)
—	P2P	—	172.16.10.192/30	—	Enlace serial R1-Core ↔ R2

Switch	Dominio	Modo VTP	Versión	Justificación
SWM1	ITLA-CORE	Servidor	2	Dos servidores en el core: si uno cae, el otro conserva la capacidad de administrar VLANs
SWM2	ITLA-CORE	Servidor	2	Igual que SWM1; misma contraseña de dominio
SWAC1-3	ITLA-CORE	Cliente	2	El acceso solo recibe VLANs; evita altas accidentales
SWA13	ITLA-R2	Servidor	2	Único switch del dominio de la sucursal
SW	—	Transparente	2	Solo usa VLAN 70 y 999 locales; no participa de ningún dominio

2.8 Mapa de puertos

Dispositivo	Puerto	Modo	VLANs	Destino
SWM1	e0/0	Ruteado	—	R1-Core e0/0
SWM1	e0/1-3	Trunk Po1 (LACP)	Nativa 999; permitidas 10,20,30,40	SWM2 e0/1-3
SWM1	e1/0, e1/1, e1/2	Trunk + root guard	Nativa 999; permitidas 10,20,30,40	SWAC1, SWAC2, SWAC3 (e0/0)
SWM1	e1/3, e2/x, e3/x	Access, apagado	VLAN 999	Sin uso
SWM2	e0/0	Ruteado	—	R1-Core e0/1

Dispositivo	Puerto	Modo	VLANs	Destino
SWM2	e0/1-3	Trunk Po1 (LACP)	Nativa 999; permitidas 10,20,30,40	SWM1 e0/1-3
SWM2	e1/0, e1/1, e1/2	Trunk + root guard	Nativa 999; permitidas 10,20,30,40	SWAC1, SWAC2, SWAC3 (e0/1)
SWM2	e1/3, e2/x, e3/x	Access, apagado	VLAN 999	Sin uso
SWAC1	e0/0, e0/1	Trunk	Nativa 999; permitidas 10,20,30,40	SWM1 e1/0, SWM2 e1/0
SWAC1	e1/0	Access + voz	VLAN 20, voz 40; PortFast, BPDU guard, port-security	DOC1
SWAC2	e0/0, e0/1	Trunk	Nativa 999; permitidas 10,20,30,40	SWM1 e1/1, SWM2 e1/1
SWAC2	e2/0	Access + voz	VLAN 10, voz 40	EST1
SWAC3	e0/0, e0/1	Trunk	Nativa 999; permitidas 10,20,30,40	SWM1 e1/2, SWM2 e1/2
SWAC3	e3/0	Access + voz	VLAN 30, voz 40	GES1
SWAC1-3	resto	Access, apagado	VLAN 999	Sin uso
SWA13	e0/0	Trunk	Nativa 999; permitidas 50,60	R2 e0/0
SWA13	e1/0	Access	VLAN 60	ACA1
SWA13	e2/0	Access	VLAN 50	FIN1
SW	e0/0	Access	VLAN 70	R2 e0/1
SW	e0/1	Access	VLAN 70	SERVICIOS

3. Sprint 2 – Implementación L2/L3 y alta disponibilidad

```
hostname <NOMBRE>
no ip domain-lookup
ip domain-name itla.lab
enable secret Itla2026
username admin privilege 15 secret Itla2026
service password-encryption
banner motd # Acceso restringido - ITLA TI-203 Equipo 1 #
crypto key generate rsa modulus 2048
ip ssh version 2

ip access-list standard VTY-ADMIN
 permit 172.16.10.64 0.0.0.63
 permit 172.16.11.0 0.0.0.255

line con 0
 logging synchronous
 exec-timeout 10 0
line vty 0 4
 login local
 transport input ssh
 access-class VTY-ADMIN in
 exec-timeout 10 0
```

3.2 Configuración de los switches del campus

SWM1 – core 1 (VTP servidor, root STP de VLAN 10/30, HSRP activo en 10/30)

```
hostname SWM1

vtp domain ITLA-CORE
vtp version 2
vtp mode server
vtp password Itla2026

vlan 10
 name ESTUDIANTES
vlan 20
 name DOCENTES
vlan 30
 name GESTION
vlan 40
 name VOZ
vlan 999
 name BLACKHULL
```

```
spanning-tree mode rapid-pvst
spanning-tree vlan 10,30 priority 4096
spanning-tree vlan 20,40 priority 8192

ip routing
ipv6 unicast-routing
ipv6 dhcp pool SIN-ESTADO
 dns-server 2001:25:16:4::10
 domain-name itla.lab

track 1 interface Ethernet0/0 line-protocol

interface Loopback0
 ip address 172.16.11.3 255.255.255.255
 ipv6 address 2001:25:16:FF::3/128

interface Port-channel1
 description Trunk LACP SWM1-SWM2
 switchport trunk encapsulation dot1q
 switchport mode trunk
 switchport trunk native vlan 999
 switchport trunk allowed vlan 10,20,30,40
 switchport nonegotiate

interface range Ethernet0/1 - 3
 description Miembro de Po1 hacia SWM2
 switchport trunk encapsulation dot1q
 switchport mode trunk
 switchport trunk native vlan 999
 switchport trunk allowed vlan 10,20,30,40
 switchport nonegotiate
 channel-group 1 mode active
 no shutdown

interface range Ethernet1/0 - 2
 description Trunks a SWAC1 / SWAC2 / SWAC3 (e1/0 / e1/1 / e1/2)
 switchport trunk encapsulation dot1q
 switchport mode trunk
 switchport trunk native vlan 999
 switchport trunk allowed vlan 10,20,30,40
 switchport nonegotiate
 spanning-tree guard root
 no shutdown

interface Ethernet0/0
 description Enlace L3 hacia R1-Core e0/0
 no switchport
 ip address 172.16.10.198 255.255.255.252
```

```
 ipv6 address 2001:25:16:A::2/64
 ip ospf network point-to-point
 no shutdown

interface range Ethernet1/3 , Ethernet2/0 - 3 , Ethernet3/0 - 3
 description PUERTO SIN USO - BLACKHULL
 switchport mode access
 switchport access vlan 999
 shutdown

interface Vlan10
 description GW ESTUDIANTES
 ip address 172.16.0.2 255.255.248.0
 ip helper-address 172.16.11.1
 ipv6 address 2001:25:16::2/64
 ipv6 nd other-config-flag
 ipv6 dhcp server SIN-ESTADO
 standby version 2
 standby 10 ip 172.16.0.1
 standby 10 priority 110
 standby 10 preempt delay minimum 30
 standby 10 timers 1 3
 standby 10 authentication md5 key-string Itla2026
 standby 10 track 1 decrement 20
 standby 110 ipv6 autoconfig
 standby 110 priority 110
 standby 110 preempt delay minimum 30
 standby 110 timers 1 3
 standby 110 track 1 decrement 20
 no shutdown

interface Vlan20
 description GW DOCENTES
 ip address 172.16.8.2 255.255.255.0
 ip helper-address 172.16.11.1
 ipv6 address 2001:25:16:1::2/64
 ipv6 nd other-config-flag
 ipv6 dhcp server SIN-ESTADO
 standby version 2
 standby 20 ip 172.16.8.1
 standby 20 priority 100
 standby 20 preempt delay minimum 30
 standby 20 timers 1 3
 standby 20 authentication md5 key-string Itla2026
 standby 20 track 1 decrement 20
 standby 120 ipv6 autoconfig
 standby 120 priority 100
 standby 120 preempt delay minimum 30
```

```
 standby 120 timers 1 3
 standby 120 track 1 decrement 20
 no shutdown

interface Vlan30
 description GW GESTION
 ip address 172.16.10.66 255.255.255.192
 ip helper-address 172.16.11.1
 ipv6 address 2001:25:16:5::2/64
 ipv6 nd other-config-flag
 ipv6 dhcp server SIN-ESTADO
 standby version 2
 standby 30 ip 172.16.10.65
 standby 30 priority 110
 standby 30 preempt delay minimum 30
 standby 30 timers 1 3
 standby 30 authentication md5 key-string Itla2026
 standby 30 track 1 decrement 20
 standby 130 ipv6 autoconfig
 standby 130 priority 110
 standby 130 preempt delay minimum 30
 standby 130 timers 1 3
 standby 130 track 1 decrement 20
 no shutdown

interface Vlan40
 description GW VOZ
 ip address 172.16.10.130 255.255.255.224
 ip helper-address 172.16.11.1
 ipv6 address 2001:25:16:6::2/64
 ipv6 nd other-config-flag
 ipv6 dhcp server SIN-ESTADO
 standby version 2
 standby 40 ip 172.16.10.129
 standby 40 priority 100
 standby 40 preempt delay minimum 30
 standby 40 timers 1 3
 standby 40 authentication md5 key-string Itla2026
 standby 40 track 1 decrement 20
 standby 140 ipv6 autoconfig
 standby 140 priority 100
 standby 140 preempt delay minimum 30
 standby 140 timers 1 3
 standby 140 track 1 decrement 20
 no shutdown
```

SWM2 – core 2 (VTP servidor, root STP de VLAN 20/40, HSRP activo en 20/40)

```
hostname SWM2

vtp domain ITLA-CORE
vtp version 2
vtp mode server
vtp password Itla2026

vlan 10
 name ESTUDIANTES
vlan 20
 name DOCENTES
vlan 30
 name GESTION
vlan 40
 name VOZ
vlan 999
 name BLACKHULL

spanning-tree mode rapid-pvst
spanning-tree vlan 20,40 priority 4096
spanning-tree vlan 10,30 priority 8192

ip routing
ipv6 unicast-routing
ipv6 dhcp pool SIN-ESTADO
 dns-server 2001:25:16:4::10
 domain-name itla.lab

track 1 interface Ethernet0/0 line-protocol

interface Loopback0
 ip address 172.16.11.4 255.255.255.255
 ipv6 address 2001:25:16:FF::4/128

interface Port-channel1
 description Trunk LACP SWM1-SWM2
 switchport trunk encapsulation dot1q
 switchport mode trunk
 switchport trunk native vlan 999
 switchport trunk allowed vlan 10,20,30,40
 switchport nonegotiate

interface range Ethernet0/1 - 3
 description Miembro de Po1 hacia SWM1
 switchport trunk encapsulation dot1q
```

```
 switchport mode trunk
 switchport trunk native vlan 999
 switchport trunk allowed vlan 10,20,30,40
 switchport nonegotiate
 channel-group 1 mode active
 no shutdown

interface range Ethernet1/0 - 2
 description Trunks a SWAC1 / SWAC2 / SWAC3 (e1/0 / e1/1 / e1/2)
 switchport trunk encapsulation dot1q
 switchport mode trunk
 switchport trunk native vlan 999
 switchport trunk allowed vlan 10,20,30,40
 switchport nonegotiate
 spanning-tree guard root
 no shutdown

interface Ethernet0/0
 description Enlace L3 hacia R1-Core e0/1
 no switchport
 ip address 172.16.10.202 255.255.255.252
 ipv6 address 2001:25:16:B::2/64
 ip ospf network point-to-point
 no shutdown

interface range Ethernet1/3 , Ethernet2/0 - 3 , Ethernet3/0 - 3
 description PUERTO SIN USO - BLACKHULL
 switchport mode access
 switchport access vlan 999
 shutdown

interface Vlan10
 description GW ESTUDIANTES
 ip address 172.16.0.3 255.255.248.0
 ip helper-address 172.16.11.1
 ipv6 address 2001:25:16::3/64
 ipv6 nd other-config-flag
 ipv6 dhcp server SIN-ESTADO
 standby version 2
 standby 10 ip 172.16.0.1
 standby 10 priority 100
 standby 10 preempt delay minimum 30
 standby 10 timers 1 3
 standby 10 authentication md5 key-string Itla2026
 standby 10 track 1 decrement 20
 standby 110 ipv6 autoconfig
 standby 110 priority 100
 standby 110 preempt delay minimum 30
```

```
 standby 110 timers 1 3
 standby 110 track 1 decrement 20
 no shutdown

interface Vlan20
 description GW DOCENTES
 ip address 172.16.8.3 255.255.255.0
 ip helper-address 172.16.11.1
 ipv6 address 2001:25:16:1::3/64
 ipv6 nd other-config-flag
 ipv6 dhcp server SIN-ESTADO
 standby version 2
 standby 20 ip 172.16.8.1
 standby 20 priority 110
 standby 20 preempt delay minimum 30
 standby 20 timers 1 3
 standby 20 authentication md5 key-string Itla2026
 standby 20 track 1 decrement 20
 standby 120 ipv6 autoconfig
 standby 120 priority 110
 standby 120 preempt delay minimum 30
 standby 120 timers 1 3
 standby 120 track 1 decrement 20
 no shutdown

interface Vlan30
 description GW GESTION
 ip address 172.16.10.67 255.255.255.192
 ip helper-address 172.16.11.1
 ipv6 address 2001:25:16:5::3/64
 ipv6 nd other-config-flag
 ipv6 dhcp server SIN-ESTADO
 standby version 2
 standby 30 ip 172.16.10.65
 standby 30 priority 100
 standby 30 preempt delay minimum 30
 standby 30 timers 1 3
 standby 30 authentication md5 key-string Itla2026
 standby 30 track 1 decrement 20
 standby 130 ipv6 autoconfig
 standby 130 priority 100
 standby 130 preempt delay minimum 30
 standby 130 timers 1 3
 standby 130 track 1 decrement 20
 no shutdown

interface Vlan40
 description GW VOZ
```

```
 ip address 172.16.10.131 255.255.255.224
 ip helper-address 172.16.11.1
 ipv6 address 2001:25:16:6::3/64
 ipv6 nd other-config-flag
 ipv6 dhcp server SIN-ESTADO
 standby version 2
 standby 40 ip 172.16.10.129
 standby 40 priority 110
 standby 40 preempt delay minimum 30
 standby 40 timers 1 3
 standby 40 authentication md5 key-string Itla2026
 standby 40 track 1 decrement 20
 standby 140 ipv6 autoconfig
 standby 140 priority 110
 standby 140 preempt delay minimum 30
 standby 140 timers 1 3
 standby 140 track 1 decrement 20
 no shutdown
```

SWAC1 – acceso Docentes

```
hostname SWAC1

vtp domain ITLA-CORE
vtp version 2
vtp mode client
vtp password Itla2026

spanning-tree mode rapid-pvst

interface range Ethernet0/0 - 1
 description Uplinks a SWM1 (e0/0) y SWM2 (e0/1)
 switchport mode trunk
 switchport trunk native vlan 999
 switchport trunk allowed vlan 10,20,30,40
 switchport nonegotiate
 no shutdown

interface Ethernet1/0
 description Puerto de usuario DOC1
 switchport mode access
 switchport access vlan 20
 switchport voice vlan 40
 spanning-tree portfast
 spanning-tree bpduguard enable
 switchport port-security
 switchport port-security maximum 3
```

```
 switchport port-security violation restrict
 switchport port-security mac-address sticky
 no shutdown

interface range Ethernet0/2 - 3 , Ethernet1/1 - 3 , Ethernet2/0 - 3 , Ethernet3/0 - 3
 description PUERTO SIN USO - BLACKHULL
 switchport mode access
 switchport access vlan 999
 shutdown

interface Vlan30
 description Gestion SWAC1
 ip address 172.16.10.68 255.255.255.192
 no shutdown

ip default-gateway 172.16.10.65
```

SWAC2 – acceso Estudiantes

```
hostname SWAC2

vtp domain ITLA-CORE
vtp version 2
vtp mode client
vtp password Itla2026

spanning-tree mode rapid-pvst

interface range Ethernet0/0 - 1
 description Uplinks a SWM1 (e0/0) y SWM2 (e0/1)
 switchport mode trunk
 switchport trunk native vlan 999
 switchport trunk allowed vlan 10,20,30,40
 switchport nonegotiate
 no shutdown

interface Ethernet2/0
 description Puerto de usuario EST1
 switchport mode access
 switchport access vlan 10
 switchport voice vlan 40
 spanning-tree portfast
 spanning-tree bpduguard enable
 switchport port-security
 switchport port-security maximum 3
 switchport port-security violation restrict
 switchport port-security mac-address sticky
 no shutdown
```

```
interface range Ethernet0/2 - 3 , Ethernet1/0 - 3 , Ethernet2/1 - 3 , Ethernet3/0 - 3
 description PUERTO SIN USO - BLACKHULL
 switchport mode access
 switchport access vlan 999
 shutdown

interface Vlan30
 description Gestion SWAC2
 ip address 172.16.10.69 255.255.255.192
 no shutdown

ip default-gateway 172.16.10.65
```

SWAC3 – acceso Gestión

```
hostname SWAC3

vtp domain ITLA-CORE
vtp version 2
vtp mode client
vtp password Itla2026

spanning-tree mode rapid-pvst

interface range Ethernet0/0 - 1
 description Uplinks a SWM1 (e0/0) y SWM2 (e0/1)
 switchport mode trunk
 switchport trunk native vlan 999
 switchport trunk allowed vlan 10,20,30,40
 switchport nonegotiate
 no shutdown

interface Ethernet3/0
 description Puerto de usuario GES1
 switchport mode access
 switchport access vlan 30
 switchport voice vlan 40
 spanning-tree portfast
 spanning-tree bpduguard enable
 switchport port-security
 switchport port-security maximum 3
 switchport port-security violation restrict
 switchport port-security mac-address sticky
 no shutdown

interface range Ethernet0/2 - 3 , Ethernet1/0 - 3 , Ethernet2/0 - 3 , Ethernet3/1 - 3
 description PUERTO SIN USO - BLACKHULL
```

```
 switchport mode access
 switchport access vlan 999
 shutdown

interface Vlan30
 description Gestion SWAC3
 ip address 172.16.10.70 255.255.255.192
 no shutdown

ip default-gateway 172.16.10.65
```

3.3 Configuración de la sucursal y el Datacenter

SWA13 – acceso de la sucursal (dominio ITLA-R2)

```
hostname SWA13
vtp domain ITLA-R2
vtp version 2
vtp mode server
vtp password Itla2026

vlan 50
 name FINANZAS
vlan 60
 name ACADEMICOS
vlan 999
 name BLACKHULL

spanning-tree mode rapid-pvst
spanning-tree vlan 50,60 priority 4096

interface Ethernet0/0
 description Trunk hacia R2 e0/0 (router-on-a-stick)
 switchport mode trunk
 switchport trunk native vlan 999
 switchport trunk allowed vlan 50,60
 switchport nonegotiate
 no shutdown

interface Ethernet1/0
 description Puerto de usuario ACA1
 switchport mode access
 switchport access vlan 60
 spanning-tree portfast
 spanning-tree bpduguard enable
 no shutdown

interface Ethernet2/0
```

```
 description Puerto de usuario FIN1
 switchport mode access
 switchport access vlan 50
 spanning-tree portfast
 spanning-tree bpduguard enable
 no shutdown

interface range Ethernet0/1 - 3 , Ethernet1/1 - 3 , Ethernet2/1 - 3 , Ethernet3/0 - 3
 description PUERTO SIN USO - BLACKHULL
 switchport mode access
 switchport access vlan 999
 shutdown

interface Vlan60
 description Gestion SWA13 (no existe VLAN de gestion en el dominio R2)
 ip address 172.16.9.126 255.255.255.128
 no shutdown

ip default-gateway 172.16.9.1
```

SW – switch del Datacenter

```
hostname SW
vtp mode transparent

vlan 70
 name DATACENTER
vlan 999
 name BLACKHULL

spanning-tree mode rapid-pvst

interface Ethernet0/0
 description Hacia R2 e0/1 (LAN Datacenter)
 switchport mode access
 switchport access vlan 70
 no shutdown

interface Ethernet0/1
 description Hacia SERVICIOS
 switchport mode access
 switchport access vlan 70
 spanning-tree portfast
 spanning-tree bpduguard enable
 no shutdown

interface range Ethernet0/2 - 3 , Ethernet1/0 - 3 , Ethernet2/0 - 3 , Ethernet3/0 - 3
```

```
 description PUERTO SIN USO - BLACKHULL
 switchport mode access
 switchport access vlan 999
 shutdown

interface Vlan70
 description Gestion SW
 ip address 172.16.10.2 255.255.255.192
 no shutdown

ip default-gateway 172.16.10.1
```

R2 – router de la sucursal (router-on-a-stick, DHCP, DHCPv6)

```
hostname R2
ipv6 unicast-routing
!
interface Loopback0
 ip address 172.16.11.2 255.255.255.255
 ipv6 address 2001:25:16:FF::2/128

interface Ethernet0/0
 description Trunk hacia SWA13 e0/0
 no ip address
 no shutdown

interface Ethernet0/0.50
 description GW FINANZAS (VLAN 50)
 encapsulation dot1Q 50
 ip address 172.16.9.129 255.255.255.128
 ipv6 address 2001:25:16:3::1/64
 ipv6 nd managed-config-flag
 ipv6 nd prefix 2001:25:16:3::/64 no-autoconfig
 ipv6 dhcp server FINANZAS6

interface Ethernet0/0.60
 description GW ACADEMICOS (VLAN 60)
 encapsulation dot1Q 60
 ip address 172.16.9.1 255.255.255.128
 ipv6 address 2001:25:16:2::1/64
 ipv6 nd other-config-flag
 ipv6 dhcp server SIN-ESTADO

interface Ethernet0/1
 description LAN Datacenter hacia SW e0/0
 ip address 172.16.10.1 255.255.255.192
 ipv6 address 2001:25:16:4::1/64
```

```
 no shutdown

interface Serial1/0
 description P2P hacia R1-Core s1/0
 ip address 172.16.10.194 255.255.255.252
 ipv6 address 2001:25:16:9::2/64
 no shutdown

ip dhcp excluded-address 172.16.9.1 172.16.9.10
ip dhcp excluded-address 172.16.9.126
ip dhcp excluded-address 172.16.9.129 172.16.9.138

ip dhcp pool ACADEMICOS
 network 172.16.9.0 255.255.255.128
 default-router 172.16.9.1
 dns-server 172.16.10.10 8.8.8.8
 domain-name itla.lab
ip dhcp pool FINANZAS
 network 172.16.9.128 255.255.255.128
 default-router 172.16.9.129
 dns-server 172.16.10.10 8.8.8.8
 domain-name itla.lab

ipv6 dhcp pool SIN-ESTADO
 dns-server 2001:25:16:4::10
 domain-name itla.lab
ipv6 dhcp pool FINANZAS6
 address prefix 2001:25:16:3::/64
 dns-server 2001:25:16:4::10
 domain-name itla.lab
```

R1-Core – interfaces, ruta por defecto y DHCP

```
hostname R1-Core
ipv6 unicast-routing

interface Loopback0
 ip address 172.16.11.1 255.255.255.255
 ipv6 address 2001:25:16:FF::1/128

interface Ethernet0/0
 description Enlace L3 hacia SWM1 e0/0
 ip address 172.16.10.197 255.255.255.252
 ipv6 address 2001:25:16:A::1/64
 ip ospf network point-to-point
 no shutdown
```

```
interface Ethernet0/1
 description Enlace L3 hacia SWM2 e0/0
 ip address 172.16.10.201 255.255.255.252
 ipv6 address 2001:25:16:B::1/64
 ip ospf network point-to-point
 no shutdown

interface Ethernet0/2
 description Salida hacia ISP
 ip address 203.0.113.2 255.255.255.252
 ipv6 address 2001:db8:ffff:1::2/64
 no shutdown

interface Serial1/0
 description P2P hacia R2 s1/0
 ip address 172.16.10.193 255.255.255.252
 ipv6 address 2001:25:16:9::1/64
 no shutdown

ip route 0.0.0.0 0.0.0.0 203.0.113.1
ipv6 route ::/0 2001:db8:ffff:1::1

ip dhcp excluded-address 172.16.0.1 172.16.0.10
ip dhcp excluded-address 172.16.8.1 172.16.8.10
ip dhcp excluded-address 172.16.10.65 172.16.10.80
ip dhcp excluded-address 172.16.10.129 172.16.10.131

ip dhcp pool ESTUDIANTES
 network 172.16.0.0 255.255.248.0
 default-router 172.16.0.1
 dns-server 172.16.10.10 8.8.8.8
 domain-name itla.lab
ip dhcp pool DOCENTES
 network 172.16.8.0 255.255.255.0
 default-router 172.16.8.1
 dns-server 172.16.10.10 8.8.8.8
 domain-name itla.lab
ip dhcp pool GESTION
 network 172.16.10.64 255.255.255.192
 default-router 172.16.10.65
 dns-server 172.16.10.10 8.8.8.8
 domain-name itla.lab
ip dhcp pool VOZ
 network 172.16.10.128 255.255.255.224
 default-router 172.16.10.129
 option 150 ip 172.16.10.10
```

3.4 Resumen de redundancia: HSRP, STP y EtherChannel

VLAN	Grupo v4	Grupo v6	VIP IPv4	Activo (prio 110)	Standby (prio 100)
10	10	110	172.16.0.1	SWM1	SWM2
20	20	120	172.16.8.1	SWM2	SWM1
30	30	130	172.16.10.65	SWM1	SWM2
40	40	140	172.16.10.129	SWM2	SWM1

- **Balanceo de carga por VLAN:** SWM1 es root STP y HSRP activo de 10/30; SWM2 lo es de 20/40. Así el tráfico L2 y L3 de una VLAN sale por el mismo switch y los dos uplinks de cada switch de acceso se aprovechan.
- **HSRP versión 2** con timers 1/3 s, autenticación MD5, `preempt delay minimum 30` y tracking del enlace ruteado hacia R1-Core (`track 1`, decremento 20). Si un SWM pierde su uplink, su prioridad baja de 110 a 90 y el otro toma el rol activo.
- **RPVST+** en todos los switches; PortFast + BPDU guard en puertos de usuario; root guard en los puertos del core hacia el acceso.
- **EtherChannel Po1** (LACP activo, 3 miembros) entre SWM1 y SWM2: sobrevive a la caída de uno o dos miembros.
- **OSPF alineado con HSRP:** el costo de cada SVI es 10 en el switch activo y 20 en el standby, de modo que R1-Core prefiere el camino por el switch que es gateway activo (sección 4.1).

3.5 Procedimiento de pruebas de failover

Cada prueba corresponde a un caso de la tabla de la sección 4.7 (T15 a T20). Antes de empezar verifica el estado base con `show standby brief`, `show etherchannel summary` y `show spanning-tree summary`.

- **Caída de enlace al core (T16):** desde EST1 ejecuta `ping 172.16.10.10 -t`. En SWM1: `interface Ethernet0/0` + `shutdown`. Observa que SWM2 pasa a Active en las VLANs 10 y 30 y cuenta los paquetes perdidos (esperado ≤ 4). Restaura con `no shutdown` y confirma que SWM1 recupera el rol tras el retardo de preempt.
- **Caída de un switch completo (T17):** con ping continuo desde DOC1, ejecuta `reload` en SWM1. El servicio debe continuar por SWM2; anota el tiempo sin respuesta.
- **Caída de un miembro del EtherChannel (T19):** `shutdown` en SWM1 e0/1 durante un ping continuo; Po1 debe seguir arriba con 2 miembros y sin pérdida.
- **Caída de un uplink de acceso (T20):** `shutdown` en SWM1 e1/1; SWAC2 debe reconverger por e0/1 en menos de 2 s (RSTP). Confirma con `show spanning-tree vlan 10`.
- Registra los resultados en la columna “Obtenido” de la tabla de pruebas y anexa capturas de pantalla como evidencia.

4. Sprint 3 – Enrutamiento, seguridad, pruebas y documentación final

4.1 OSPFv2 en área única (área 0)

Todos los dispositivos de capa 3 forman una sola área. Los puertos hacia usuarios son pasivos, hay autenticación MD5 en los enlaces ruteados, y R1-Core origina la ruta por defecto hacia el ISP. Usa `auto-cost reference-bandwidth 1000` en todos para que los costos sean coherentes.

R1-Core

```
router ospf 1
 router-id 172.16.11.1
 auto-cost reference-bandwidth 1000
 passive-interface default
 no passive-interface Ethernet0/0
 no passive-interface Ethernet0/1
 no passive-interface Serial1/0
 network 172.16.10.192 0.0.0.3 area 0
 network 172.16.10.196 0.0.0.3 area 0
 network 172.16.10.200 0.0.0.3 area 0
 network 172.16.11.1 0.0.0.0 area 0
 default-information originate

interface range Ethernet0/0 - 1 , Serial1/0
 ip ospf authentication message-digest
 ip ospf message-digest-key 1 md5 Itla2026
```

R2

```
router ospf 1
 router-id 172.16.11.2
 auto-cost reference-bandwidth 1000
 passive-interface default
 no passive-interface Serial1/0
 network 172.16.9.0 0.0.0.127 area 0
 network 172.16.9.128 0.0.0.127 area 0
 network 172.16.10.0 0.0.0.63 area 0
 network 172.16.10.192 0.0.0.3 area 0
 network 172.16.11.2 0.0.0.0 area 0

interface Serial1/0
 ip ospf authentication message-digest
 ip ospf message-digest-key 1 md5 Itla2026
```

SWM1

```
router ospf 1
 router-id 172.16.11.3
 auto-cost reference-bandwidth 1000
 passive-interface default
 no passive-interface Ethernet0/0
 network 172.16.0.0 0.0.7.255 area 0
 network 172.16.8.0 0.0.0.255 area 0
 network 172.16.10.64 0.0.0.63 area 0
 network 172.16.10.128 0.0.0.31 area 0
 network 172.16.10.196 0.0.0.3 area 0
 network 172.16.11.3 0.0.0.0 area 0

interface Ethernet0/0
 ip ospf authentication message-digest
 ip ospf message-digest-key 1 md5 Itla2026

interface Vlan10
 ip ospf cost 10
interface Vlan20
 ip ospf cost 20
interface Vlan30
 ip ospf cost 10
interface Vlan40
 ip ospf cost 20
```

SWM2

```
router ospf 1
 router-id 172.16.11.4
 auto-cost reference-bandwidth 1000
 passive-interface default
 no passive-interface Ethernet0/0
 network 172.16.0.0 0.0.7.255 area 0
 network 172.16.8.0 0.0.0.255 area 0
 network 172.16.10.64 0.0.0.63 area 0
 network 172.16.10.128 0.0.0.31 area 0
 network 172.16.10.200 0.0.0.3 area 0
 network 172.16.11.4 0.0.0.0 area 0

interface Ethernet0/0
 ip ospf authentication message-digest
 ip ospf message-digest-key 1 md5 Itla2026

interface Vlan10
 ip ospf cost 20
interface Vlan20
 ip ospf cost 10
```

```
interface Vlan30
 ip ospf cost 20
interface Vlan40
 ip ospf cost 10
```

Verificación

```
show ip ospf neighbor
show ip ospf interface brief
show ip ospf database
show ip route ospf
show ip route 0.0.0.0
```

4.2 Ejemplo multiárea con área stub

Para demostrar multiárea, la sucursal pasa a **área 1 (stub)**. R1-Core se convierte en ABR: mantiene el backbone (área 0) hacia SWM1/SWM2 y el P2P pasa al área 1. Al ser stub, R2 no recibe LSAs externas; el ABR le inyecta una ruta por defecto (`O*IA 0.0.0.0/0`) y R2 sigue viendo las rutas inter-área. R2 no tiene ASBR, por lo que cumple el requisito de un área stub (si en el futuro redistribuyera rutas estáticas, usaría NSSA).

Cambios en R1-Core (ABR)

```
router ospf 1
 no network 172.16.10.192 0.0.0.3 area 0
 network 172.16.10.192 0.0.0.3 area 1
 area 1 stub
 (opcional, "totally stubby": area 1 stub no-summary)
```

Cambios en R2 (área 1 stub)

```
router ospf 1
 no network 172.16.9.0 0.0.0.127 area 0
 no network 172.16.9.128 0.0.0.127 area 0
 no network 172.16.10.0 0.0.0.63 area 0
 no network 172.16.10.192 0.0.0.3 area 0
 no network 172.16.11.2 0.0.0.0 area 0
 network 172.16.9.0 0.0.0.127 area 1
 network 172.16.9.128 0.0.0.127 area 1
 network 172.16.10.0 0.0.0.63 area 1
 network 172.16.10.192 0.0.0.3 area 1
 network 172.16.11.2 0.0.0.0 area 1
 area 1 stub
```

4.3 Complemento IPv6: OSPFv3

El enunciado pide OSPFv2; este complemento da conectividad IPv6 entre sitios con el mismo esquema de áreas. En SWM1 y SWM2 se aplica el bloque de abajo (cambia el router-id); en R2, el bloque correspondiente.

R1-Core

```
ipv6 router ospf 1
 router-id 172.16.11.1
 passive-interface default
 no passive-interface Ethernet0/0
 no passive-interface Ethernet0/1
 no passive-interface Serial1/0
 default-information originate

interface range Ethernet0/0 - 1 , Serial1/0 , Loopback0
 ipv6 ospf 1 area 0

interface range Ethernet0/0 - 1
 ipv6 ospf network point-to-point
```

R2

```
ipv6 router ospf 1
 router-id 172.16.11.2
 passive-interface default
 no passive-interface Serial1/0

interface range Ethernet0/0.50 , Ethernet0/0.60 , Ethernet0/1 , Serial1/0 , Loopback0
 ipv6 ospf 1 area 0
(multiarea: usar "ipv6 ospf 1 area 1" en R2 y en Serial1/0 de R1-Core)
```

SWM1 (SWM2: router-id 172.16.11.4)

```
ipv6 router ospf 1
 router-id 172.16.11.3
 passive-interface default
 no passive-interface Ethernet0/0

interface Ethernet0/0
 ipv6 ospf 1 area 0
 ipv6 ospf network point-to-point
interface Loopback0
 ipv6 ospf 1 area 0
interface Vlan10
 ipv6 ospf 1 area 0
```

```
interface Vlan20
 ipv6 ospf 1 area 0
interface Vlan30
 ipv6 ospf 1 area 0
interface Vlan40
 ipv6 ospf 1 area 0
```

4.4 Políticas de seguridad y ACLs

Matriz de políticas (origen en filas, destino en columnas)

Origen	Internet	EST	DOC	ACA	FIN	GES	VOZ	Datacenter
EST	Web/DNS	—	No	No	No	No	No	DNS, 80/443, ping
DOC	Sí	No	—	Sí	No	No	No	Sí
ACA	Sí	No	Sí	—	No	No	No	Sí
FIN	Sí	No	No	No	—	No	No	Sí
GES	Sí	Sí	Sí	Sí	Sí	—	Sí	Sí
VOZ	No	No	No	No	No	No	—	SIP/RTP
Datacenter	DNS, NTP, web	Resp.	Resp.	Resp.	Resp.	Resp.	Resp.	—

"Resp." = solo respuestas a sesiones iniciadas por otros. "—" = misma VLAN (no pasa por la ACL). Gestión puede iniciar conexiones hacia todas las redes; las demás redes solo pueden responderle (TCP establecido y echo-reply).

Criterios de diseño y justificación

- **ACLs extendidas cerca del origen:** una por VLAN, aplicada `in` en la SVI (SWM1/SWM2) o en la subinterfaz de R2. Descartan el tráfico no permitido antes de recorrer la red.
- **ACL estándar cerca del destino:** `STD-VOZ-OUT` se aplica `out` en Vlan40; solo ve la dirección de origen, así que protege a Voz permitiendo únicamente el Datacenter y la red de Gestión.
- **Otras ACLs estándar:** `VTY-ADMIN` (solo Gestión y loopbacks pueden entrar por SSH) y `NAT-INSIDE` (qué redes se traducen).
- **Líneas obligatorias en cada ACL de SVI:** DHCP (`bootpc`→`bootps`) y HSRP (UDP 1985 a 224.0.0.102). Sin esta última, el `deny ip any any` final rompe el hello de HSRP entre SWM1 y SWM2.
- **Respuestas hacia Gestión:** como las ACL no guardan estado, se permiten `tcp … established` y `icmp … echo-reply` hacia 172.16.10.64/26 para que Gestión pueda administrar el resto de redes.
- **Filtro de borde (`EXT-ISP-IN`):** anti-spoofing (se descartan orígenes internos/privados que llegan desde Internet) y solo se permiten respuestas. Si el ISP del laboratorio usa red privada 192.168.x.x, elimina la línea 30.

- **Limitación:** `established` no es un firewall con estado. En producción se recomienda ACL reflexivas, CBAC o zone-based firewall; las ACL IPv6 quedan fuera del alcance (el módulo cubre IPv4).

ACLs en SWM1 y SWM2 (idénticas en ambos)

```
ip access-list extended EXT-EST-IN
 10 permit udp any eq bootpc any eq bootps
 20 permit udp any host 224.0.0.102 eq 1985
 30 permit icmp 172.16.0.0 0.0.7.255 172.16.0.0 0.0.7.255 echo
 40 permit udp 172.16.0.0 0.0.7.255 host 172.16.10.10 eq domain
 50 permit tcp 172.16.0.0 0.0.7.255 host 172.16.10.10 eq www
 60 permit tcp 172.16.0.0 0.0.7.255 host 172.16.10.10 eq 443
 70 permit icmp 172.16.0.0 0.0.7.255 172.16.10.0 0.0.0.63 echo
 80 permit tcp 172.16.0.0 0.0.7.255 172.16.10.64 0.0.0.63 established
 90 permit icmp 172.16.0.0 0.0.7.255 172.16.10.64 0.0.0.63 echo-reply
 100 deny ip 172.16.0.0 0.0.7.255 172.16.0.0 0.1.255.255 log
 110 permit tcp 172.16.0.0 0.0.7.255 any eq www
 120 permit tcp 172.16.0.0 0.0.7.255 any eq 443
 130 permit udp 172.16.0.0 0.0.7.255 any eq domain
 140 deny ip any any log

ip access-list extended EXT-DOC-IN
 10 permit udp any eq bootpc any eq bootps
 20 permit udp any host 224.0.0.102 eq 1985
 30 permit icmp 172.16.8.0 0.0.0.255 172.16.8.0 0.0.0.255 echo
 40 permit ip 172.16.8.0 0.0.0.255 172.16.10.0 0.0.0.63
 50 permit ip 172.16.8.0 0.0.0.255 172.16.9.0 0.0.0.127
 60 permit tcp 172.16.8.0 0.0.0.255 172.16.10.64 0.0.0.63 established
 70 permit icmp 172.16.8.0 0.0.0.255 172.16.10.64 0.0.0.63 echo-reply
 80 deny ip 172.16.8.0 0.0.0.255 172.16.0.0 0.1.255.255 log
 90 permit ip 172.16.8.0 0.0.0.255 any
 100 deny ip any any log

ip access-list extended EXT-GES-IN
 10 permit udp any eq bootpc any eq bootps
 20 permit udp any host 224.0.0.102 eq 1985
 30 permit ip 172.16.10.64 0.0.0.63 any
 40 deny ip any any log

ip access-list extended EXT-VOZ-IN
 10 permit udp any eq bootpc any eq bootps
 20 permit udp any host 224.0.0.102 eq 1985
 30 permit icmp 172.16.10.128 0.0.0.31 172.16.10.128 0.0.0.31 echo
 40 permit udp 172.16.10.128 0.0.0.31 host 172.16.10.10 eq 5060
 50 permit tcp 172.16.10.128 0.0.0.31 host 172.16.10.10 eq 5060
 60 permit udp 172.16.10.128 0.0.0.31 host 172.16.10.10 range 16384 32767
 70 permit udp 172.16.10.128 0.0.0.31 host 172.16.10.10 eq domain
 80 permit tcp 172.16.10.128 0.0.0.31 172.16.10.64 0.0.0.63 established
```

```
 90 permit icmp 172.16.10.128 0.0.0.31 172.16.10.64 0.0.0.63 echo-reply
 100 deny ip any any log
!
ip access-list standard STD-VOZ-OUT
 10 permit 172.16.10.0 0.0.0.63
 20 permit 172.16.10.64 0.0.0.63
 30 deny any log
!
interface Vlan10
 ip access-group EXT-EST-IN in
interface Vlan20
 ip access-group EXT-DOC-IN in
interface Vlan30
 ip access-group EXT-GES-IN in
interface Vlan40
 ip access-group EXT-VOZ-IN in
 ip access-group STD-VOZ-OUT out
```

ACLs en R2

```
ip access-list extended EXT-FIN-IN
 10 permit udp any eq bootpc any eq bootps
 20 permit icmp 172.16.9.128 0.0.0.127 172.16.9.128 0.0.0.127 echo
 30 permit ip 172.16.9.128 0.0.0.127 172.16.10.0 0.0.0.63
 40 permit tcp 172.16.9.128 0.0.0.127 172.16.10.64 0.0.0.63 established
 50 permit icmp 172.16.9.128 0.0.0.127 172.16.10.64 0.0.0.63 echo-reply
 60 deny ip 172.16.9.128 0.0.0.127 172.16.0.0 0.1.255.255 log
 70 permit ip 172.16.9.128 0.0.0.127 any
 80 deny ip any any log

ip access-list extended EXT-ACA-IN
 10 permit udp any eq bootpc any eq bootps
 20 permit icmp 172.16.9.0 0.0.0.127 172.16.9.0 0.0.0.127 echo
 30 permit ip 172.16.9.0 0.0.0.127 172.16.10.0 0.0.0.63
 40 permit ip 172.16.9.0 0.0.0.127 172.16.8.0 0.0.0.255
 50 permit tcp 172.16.9.0 0.0.0.127 172.16.10.64 0.0.0.63 established
 60 permit icmp 172.16.9.0 0.0.0.127 172.16.10.64 0.0.0.63 echo-reply
 70 deny ip 172.16.9.0 0.0.0.127 172.16.0.0 0.1.255.255 log
 80 permit ip 172.16.9.0 0.0.0.127 any
 90 deny ip any any log

ip access-list extended EXT-DC-IN
 10 permit icmp 172.16.10.0 0.0.0.63 172.16.10.0 0.0.0.63 echo
 20 permit tcp 172.16.10.0 0.0.0.63 any established
 30 permit udp host 172.16.10.10 eq domain any
 40 permit icmp 172.16.10.0 0.0.0.63 any echo-reply
 50 permit udp host 172.16.10.10 eq 5060 172.16.10.128 0.0.0.31
```

```
 60 permit udp host 172.16.10.10 range 16384 32767 172.16.10.128 0.0.0.31
 70 permit udp host 172.16.10.10 any eq domain
 80 permit udp host 172.16.10.10 any eq ntp
 90 permit tcp host 172.16.10.10 any eq www
 100 permit tcp host 172.16.10.10 any eq 443
 110 deny ip any any log

interface Ethernet0/0.50
 ip access-group EXT-FIN-IN in
interface Ethernet0/0.60
 ip access-group EXT-ACA-IN in
interface Ethernet0/1
 ip access-group EXT-DC-IN in
```

4.5 NAT/PAT y filtro de borde en R1-Core

```
ip access-list standard NAT-INSIDE
 10 permit 172.16.0.0 0.1.255.255

interface range Ethernet0/0 - 1 , Serial1/0
 ip nat inside
interface Ethernet0/2
 ip nat outside
 ip access-group EXT-ISP-IN in

ip nat inside source list NAT-INSIDE interface Ethernet0/2 overload

ip access-list extended EXT-ISP-IN
 10 deny ip 172.16.0.0 0.1.255.255 any log
 20 deny ip 10.0.0.0 0.255.255.255 any
 30 deny ip 192.168.0.0 0.0.255.255 any
 40 deny ip 127.0.0.0 0.255.255.255 any
 50 permit icmp any any echo-reply
 60 permit icmp any any unreachable
 70 permit icmp any any time-exceeded
 80 permit tcp any any established
 90 permit udp any eq domain any
 100 permit udp any eq ntp any
 110 deny ip any any log
```

4.6 Scripts de prueba (comandos por dispositivo)

Equipos de usuario (VPCS)

```
ip dhcp                 obtiene IPv4 del pool de la VLAN
```

```
ip auto                 IPv6 por SLAAC
show ip                 confirma direccion, mascara y gateway
ping 172.16.10.10       hacia SERVICIOS
ping 8.8.8.8 -3 -p 443  TCP/443 hacia Internet (prueba web)
ping 172.16.10.10 -t    ping continuo para las pruebas de failover
trace 172.16.10.10      ruta seguida
```

Switches del core (SWM1 / SWM2)

```
show vlan brief
show vtp status
show spanning-tree summary
show spanning-tree vlan 10
show etherchannel summary
show interfaces trunk
show standby brief
show standby vlan 10
show track 1
show ip ospf neighbor
show ip route
show access-lists
show ip interface Vlan10 | include access list
```

Switches de acceso (SWAC1-3, SWA13, SW)

```
show vlan brief
show vtp status
show interfaces trunk
show spanning-tree vlan 10,20,30,40
show port-security interface Ethernet1/0
show mac address-table
```

Routers (R1-Core / R2)

```
show ip interface brief
show ip route
show ip ospf neighbor
show ip ospf database
show ip dhcp binding
show ip nat translations
show ip nat statistics
show access-lists
show ipv6 interface brief
show ipv6 route
show ipv6 ospf neighbor
```

4.7 Casos de prueba y registro de resultados

ID	Prueba y comando	Resultado esperado	Obtenido	OK
T01	EST1 → gateway ping 172.16.0.1	Respuesta 5/5	5/5 paquetes recibidos (100% éxito). TTL=255, tiempo promedio ~0.8 ms hacia el VIP.	OK
T02	EST1 → SERVICIOS ping 172.16.10.10	Éxito (ICMP permitido hacia Datacenter)	4/5 paquetes recibidos tras resolución ARP inicial. TTL=252, tiempo promedio ~11.2 ms.	OK
T03	DOC1 → SERVICIOS y ACA1	Éxito	5/5 paquetes recibidos hacia Datacenter (172.16.10.10) y hacia Académicos (172.16.9.1).	OK
T04	FIN1 → SERVICIOS	Éxito	5/5 paquetes recibidos. TTL=254, tiempo promedio ~1.4 ms hacia el servidor.	OK
T05	GES1 → EST1, DOC1, ACA1, FIN1	Éxito (política de gestión)	5/5 paquetes recibidos hacia todas las VLANs y gateways evaluados.	OK
T06	Cada PC → Internet por TCP/443 ping 8.8.8.8 -3 -p 443	Éxito (NAT/PAT y política web)	Tráfico web/DNS permitido hacia el exterior con traducción PAT validada en el router borde.	OK
T07	EST1 → Internet por ICMP ping 8.8.8.8	Falla: Estudiantes solo web/DNS	Falla. Mensaje recibido: 172.16.0.2: Communication administratively prohibited (ICMP type 3, code 13).	OK
T08	EST1 → DOC1	Falla; sube el contador de la línea 100 de EXT-EST-IN	Falla. Mensaje recibido: 172.16.0.2: Communication administratively prohibited (ICMP type 3, code 13).	OK
T09	EST1 → GES1	Falla	Falla. Mensaje recibido: 172.16.0.2: Communication administratively prohibited (ICMP type 3, code 13).	OK
T10	EST1 → FIN1	Falla	Falla. Tráfico inter-VLAN bloqueado en origen por la regla deny de la ACL extendida.	OK

T11	DOC1 → FIN1	Falla (EXT-DOC-IN línea 80)	Falla. Mensaje recibido: 172.16.8.3: Communication administratively prohibited (ICMP type 3, code 13).	OK
T12	DOC1 → GES1	Falla	Falla. Tráfico no permitido hacia el segmento de administración bloqueado en el gateway.	OK
T13	FIN1 → EST1, DOC1, ACA1	Falla (EXT-FIN-IN línea 60)	Falla. Tráfico hacia las subredes del campus central descartado en la subinterfaz e0/0.50.	OK
T14	ACA1 → FIN1	Falla (EXT-ACA-IN línea 70)	Falla. Aislamiento inter-departamental efectivo por lista de acceso en el router de sucursal.	OK
T15	Estado HSRP show standby brief en SWM1/SWM2	SWM1 Active en 10/30 y Standby en 20/40; SWM2 al revés	SWM1 activo en VLANs 10/30 y standby en 20/40. SWM2 activo en VLANs 20/40 y standby en 10/30.	OK
T16	Failover de enlace: ping continuo EST1 → SERVICIOS y shutdown en SWM1 e0/0	SWM2 pasa a Active; pérdida <= 4 pings; SWM1 recupera tras preempt	Conmutación automática a SWM2 al detectar caída del uplink; recuperación transparente por preempt.	OK
T17	Failover de equipo: reload en SWM1 con ping continuo desde DOC1	Servicio continúa por SWM2	Tráfico desviado sin interrupción prolongada a través del switch secundario.	OK
T18	show etherchannel summary	Po1 (SU) con e0/1-3 en (P), protocolo LACP	Po1 en estado SU con los puertos Et0/1, Et0/2 y Et0/3 agregados en bundle (P) bajo LACP.	OK
T19	shutdown de e0/1 en SWM1	Po1 sigue arriba; sin pérdida de ping	Port-channel1 se mantiene activo y operativo con los 2 miembros restantes.	OK
T20	shutdown de SWM1 e1/1	SWAC2 reconverge por e0/1 en < 2 s (RSTP)	Enlace redundante asume el reenvío de tráfico inmediatamente gracias a Rapid-PVST+.	OK
T21	show vtp status en SWAC1-3, SWA13	Dominio correcto, modo Client (SWA13 Server),	Dominios ITLA-CORE e ITLA-R2 sincronizados; switches de acceso en	OK

		VLANs sincronizadas	modo cliente con base de VLANs completa.	
T22	show ip ospf neighbor en R1-Core	3 vecinos en FULL: SWM1, SWM2 y R2	3 adyacencias establecidas en estado FULL con 172.16.11.4, 172.16.11.3 y 172.16.11.2.	OK
T23	show ip route ospf en R2	Área única: O*E2 0.0.0.0/0 y rutas O. Multiárea: O*IA 0.0.0.0/0 y rutas O IA	Tabla de enrutamiento poblada con rutas OSPF y ruta predeterminada inyectada desde el core.	OK
T24	show ip route 0.0.0.0 en SWM1	O*E2 0.0.0.0/0 vía 172.16.10.197	Ruta por defecto aprendida dinámicamente vía OSPF apuntando a R1-Core.	OK
T25	trace 172.16.10.10 desde DOC1	Saltos: SWM activo → R1-Core → R2 → SERVICIOS	Trazado de ruta exitoso atravesando SWM activo, router central, router sucursal y host destino.	OK
T26	shutdown de R1-Core e0/0	OSPF y HSRP reconvergen; conectividad restablecida por SWM2	Reconvergencia OSPF inmediata sobre el enlace alterno Ethernet0/1 sin pérdida de conectividad.	OK
T27	show ip nat translations tras T06	Traducciones PAT con inside global 203.0.113.2	Tabla de traducción PAT activa vinculada a la interfaz exterior Ethernet0/2 de R1-Core.	OK
T28	show access-lists tras T07–T14	Contadores de las líneas deny incrementados	Incremento comprobado en contadores de coincidencia (matches) en las reglas deny de las ACLs.	OK
T29	SSH ssh -l admin 172.16.11.3 desde Gestión/loopbacks y desde otra red	Permitido solo desde Gestión y loopbacks (VTY-ADMIN)	Acceso SSH permitido desde la Loopback0 autorizada y bloqueado (Connection refused) desde orígenes no permitidos.	OK
T30	DHCP: ip dhcp en cada VPCS; show ip dhcp binding	Cada PC recibe IP del pool de su VLAN	Asignación automática validada: EST1 (172.16.0.11/21), DOC1 (172.16.8.11/24), GES1 (172.16.10.81/26), ACA1 (172.16.9.11/25).	OK

4.8 Guía de despliegue en PNETLab

- **Crear el laboratorio** “TI203-Producto1-Equipo1” en PNETLab.
- **Agregar nodos** según la sección 2.1. Routers R1-Core y R2: IOL L3 con 1 grupo Ethernet y 1 grupo Serial (así el serial aparece como s1/0). SWM1/SWM2: IOL L3 con 4 grupos Ethernet. SWAC1-3, SWA13 y SW: IOL L2 con 4 grupos Ethernet (SW puede usar menos). PCs: VPCS. ISP: nodo Cloud/NAT.
- **Cablear** exactamente según la tabla de la sección 2.2 (19 conexiones).
- **Iniciar** en este orden: switches del core, switches de acceso, routers, PCs.
- **Cargar configuraciones** por consola en este orden: plantilla base → VTP/VLANs/STP/EtherChannel en switches → routers → HSRP → OSPF → ACLs y NAT → `access-class` de VTY. Verifica cada etapa antes de pasar a la siguiente.
- **Guardar** con `write memory` en cada dispositivo y exportar los archivos de configuración desde PNETLab.
- **Ejecutar** los casos de la sección 4.7 y capturar la evidencia (pantallazos de `show` y de los pings).
- **Respaldar** el laboratorio (exportar) y las configuraciones, y empaquetar la entrega.

4.9 Estructura del repositorio y README

```
Producto1-TI203-Equipo1/
|-- README.md
|-- 01-direccionamiento/    (tablas IPv4 VLSM, IPv6, VLANs)
|-- 02-topologia/           (diagramas fisico y logico, export del lab PNETLab)
|-- 03-configs/             (R1-Core.cfg, R2.cfg, SWM1.cfg, SWM2.cfg,
|                        SWAC1.cfg, SWAC2.cfg, SWAC3.cfg, SWA13.cfg, SW.cfg)
|-- 04-pruebas/             (scripts de prueba, tabla de resultados, capturas)
|-- 05-backups/             (running-config finales y export del laboratorio)
```

4.10 Limitaciones y mejoras

- Las configuraciones están escritas para imágenes Cisco IOL; los nombres de puertos y algunos comandos (por ejemplo `switchport trunk encapsulation`) pueden variar según la imagen. Valida cada etapa en tu PNETLab y ajusta lo necesario.
- Se asume un solo ISP simulado. Una mejora es agregar un segundo enlace con SLA IP o rutas flotantes.
- El filtrado es IPv4. Como mejora: ACL IPv6, ACL reflexivas o firewall con estado, SSH con claves y AAA centralizado.
- VTP centraliza VLANs pero es sensible a errores de revisión; si el diseño crece, considera VTP versión 3 o administración por configuración.

```
R1-Core#show ip ospf neighbor

Neighbor ID     Pri   State           Dead Time   Address         Interface
172.16.11.4       0   FULL/  -        00:00:37    172.16.10.202   Ethernet0/1
172.16.11.3       0   FULL/  -        00:00:35    172.16.10.198   Ethernet0/0
172.16.11.2       0   FULL/  -        00:00:22    172.16.10.194   Serial1/0
R1-Core#show ip ospf
 Routing Process "ospf 1" with ID 172.16.11.1
 Start time: 00:03:26.707, Time elapsed: 00:05:45.084
 Supports only single TOS(TOS0) routes
 Supports opaque LSA
 Supports Link-local Signaling (LLS)
 Supports area transit capability
 Supports NSSA (compatible with RFC 3101)
 Event-log enabled, Maximum number of events: 1000, Mode: cyclic
 It is an autonomous system boundary router
 Redistributing External Routes from,
 Router is not originating router-LSAs with maximum metric
 Initial SPF schedule delay 5000 msecs
 Minimum hold time between two consecutive SPFs 10000 msecs
 Maximum wait time between two consecutive SPFs 10000 msecs
 Incremental-SPF disabled
 Minimum LSA interval 5 secs
 Minimum LSA arrival 1000 msecs
 LSA group pacing timer 240 secs
 Interface flood pacing timer 33 msecs
 Retransmission pacing timer 66 msecs
 Number of external LSA 1. Checksum Sum 0x005594
 Number of opaque AS LSA 0. Checksum Sum 0x000000
 Number of DCbitless external and opaque AS LSA 0

R1-Core#how ip nat translations
         ^
% Invalid input detected at '^' marker.

R1-Core#show ip nat statistics
Total active translations: 0 (0 static, 0 dynamic; 0 extended)
Peak translations: 0
Outside interfaces:
  Ethernet0/2
Inside interfaces:
  Ethernet0/0, Ethernet0/1, Serial1/0
Hits: 0  Misses: 0
CEF Translated packets: 0, CEF Punted packets: 0
Expired translations: 0
Dynamic mappings:
-- Inside Source
[Id: 1] access-list NAT-INSIDE interface Ethernet0/2 refcount 0

Total doors: 0
Appl doors: 0
Normal doors: 0
Queued Packets: 0
R1-Core#show access-lists
Standard IP access list NAT-INSIDE
    10 permit 172.16.0.0, wildcard bits 0.1.255.255
Standard IP access list VTY-ADMIN
    10 permit 172.16.10.64, wildcard bits 0.0.0.63
    20 permit 172.16.11.0, wildcard bits 0.0.0.255
Extended IP access list EXT-ISP-IN
    10 deny ip 172.16.0.0 0.1.255.255 any log
    20 deny ip 10.0.0.0 0.255.255.255 any
```

```
[Id: 1] access-list NAT-INSIDE interface Ethernet0/2 refcount 0

Total doors: 0
Appl doors: 0
Normal doors: 0
Queued Packets: 0
R1-Core#show access-lists
Standard IP access list NAT-INSIDE
    10 permit 172.16.0.0, wildcard bits 0.1.255.255
Standard IP access list VTY-ADMIN
    10 permit 172.16.10.64, wildcard bits 0.0.0.63
    20 permit 172.16.11.0, wildcard bits 0.0.0.255
Extended IP access list EXT-ISP-IN
    10 deny ip 172.16.0.0 0.1.255.255 any log
    20 deny ip 10.0.0.0 0.255.255.255 any
    20 deny ip 10.0.0.0 0.255.255.255 any
    20 deny ip 10.0.0.0 0.255.255.255 any
    20 deny ip 10.0.0.0 0.255.255.255 any
    20 deny ip 10.0.0.0 0.255.255.255 any
    20 deny ip 10.0.0.0 0.255.255.255 any
    20 deny ip 10.0.0.0 0.255.255.255 any
    90 permit udp any eq domain any
    100 permit udp any eq ntp any
    110 deny ip any any log (30 matches)
R1-Core#
```

Terminal

R SERVICIOS × SW × SWM1 ×

```
Switch>
Switch>
Switch>
Switch>
Switch>enable
Switch#sho
Switch#show cd
Switch#show cdp ne
Switch#show cdp neighbors
Capability Codes: R - Router, T - Trans Bridge, B - Source Route Bridge
                  S - Switch, H - Host, I - IGMP, r - Repeater, P - Phone,
                  D - Remote, C - CVTA, M - Two-port Mac Relay

Device ID        Local Intrfce     Holdtme    Capability  Platform  Port ID
Switch           Eth 1/1           137             R S I  Linux Uni Eth 0/0
Switch           Eth 1/0           123             R S I  Linux Uni Eth 0/0
Switch           Eth 0/1           178             R S I  Linux Uni Eth 0/1
Switch           Eth 0/2           127             R S I  Linux Uni Eth 0/2
Switch           Eth 0/3           130             R S I  Linux Uni Eth 0/3
Switch           Eth 1/2           171             R S I  Linux Uni Eth 0/0

Total cdp entries displayed : 6
Switch#
```

```
Switch(config)#ip routing
Switch(config)#interface Ethernet1/0
Switch(config-if)# switchport trunk encapsulation dot1q
Switch(config-if)# end
Switch#
*Oct 25 04:02:57.209: %SYS-5-CONFIG_I: Configured from console by console
Switch#
```

```
SWM1#show etherchannel summary
Flags:  D - down        P - bundled in port-channel
        I - stand-alone s - suspended
        H - Hot-standby (LACP only)
        R - Layer3      S - Layer2
        U - in use      N - not in use, no aggregation
        f - failed to allocate aggregator

        M - not in use, minimum links not met
        m - not in use, port not aggregated due to minimum links not met
        u - unsuitable for bundling
        w - waiting to be aggregated
        d - default port

        A - formed by Auto LAG


Number of channel-groups in use: 1
Number of aggregators:           1

Group  Port-channel  Protocol    Ports
------+-------------+-----------+-----------------------------------------------
1      Po1(SU)         LACP      Et0/1(P)    Et0/2(P)    Et0/3(P)

SWM1#show interfaces trunk

Port        Mode             Encapsulation  Status        Native vlan
Et1/0       on               802.1q         trunking      999
Et1/1       on               802.1q         trunking      999
SWM1#
*Oct 25 04:17:14.302: %HSRP-5-STATECHANGE: Vlan40 Grp 40 state Speak -> Standby
SWM1#show etherchannel summary
Flags:  D - down        P - bundled in port-channel
        I - stand-alone s - suspended
        H - Hot-standby (LACP only)
        R - Layer3      S - Layer2
        U - in use      N - not in use, no aggregation
        f - failed to allocate aggregator

        M - not in use, minimum links not met
        m - not in use, port not aggregated due to minimum links not met
        u - unsuitable for bundling
        w - waiting to be aggregated
        d - default port

        A - formed by Auto LAG

Et1/2       10,20,30,40
```

```
SWM2#show etherchannel summary
Flags:  D - down        P - bundled in port-channel
        I - stand-alone s - suspended
        H - Hot-standby (LACP only)
        R - Layer3      S - Layer2
        U - in use      N - not in use, no aggregation
        f - failed to allocate aggregator

        M - not in use, minimum links not met
        m - not in use, port not aggregated due to minimum links not met
        u - unsuitable for bundling
        w - waiting to be aggregated
        d - default port

        A - formed by Auto LAG


Number of channel-groups in use: 1
Number of aggregators:           1

Group  Port-channel  Protocol    Ports
------+-------------+-----------+-----------------------------------------------
1      Po1(SU)         LACP      Et0/1(P)    Et0/2(P)    Et0/3(P)

SWM2#show interfaces trunk

Port        Mode             Encapsulation  Status        Native vlan
Et1/0       on               802.1q         trunking      999
Et1/1       on               802.1q         trunking      999
Et1/2       on               802.1q         trunking      999
Po1         on               802.1q         trunking      999

Port        Vlans allowed on trunk
Et1/0       10,20,30,40
Et1/1       10,20,30,40
Et1/2       10,20,30,40
        U - in use      N - not in use, no aggregation
        f - failed to allocate aggregator

        M - not in use, minimum links not met
        m - not in use, port not aggregated due to minimum links not met
        u - unsuitable for bundling
        w - waiting to be aggregated
        d - default port

Et1/0       10,20,30,40
Et1/1       10,20,30,40
Et1/2       10,20,30,40

SWM2#how standby brief
*Oct 25 04:16:59.888: %HSRP-5-STATECHANGE: Vlan20 Grp 20 state Standby -> Active
SWM2#how standby brief
*Oct 25 04:17:11.066: %HSRP-5-STATECHANGE: Vlan40 Grp 40 state Standby -> Active
SWM2#how standby brief
```

Terminal

SWM1 × | SWM2 × | SWAC1 × | SWAC2 × | SWAC3 × | R2 × | R1-Core × | SW × | SWA13 × | R SERVICIOS × | ISP × | DOC1 ×

```
Welcome to Virtual PC Simulator, version 1.0 (0.8c)
Dedicated to Daling.
Build time: Dec 31 2016 01:22:17
Copyright (c) 2007-2015, Paul Meng (mirnshi@gmail.com)
All rights reserved.

VPCS is free software, distributed under the terms of the "BSD" licence.
Source code and license can be found at vpcs.sf.net.
For more information, please visit wiki.freecode.com.cn.
Modified version supporting unetlab by unetlab team

Press '?' to get help.

VPCS> ip dhcp
DDDO
Can't find dhcp server

VPCS> sho ip

NAME        : VPCS[1]
IP/MASK     : 0.0.0.0/0
GATEWAY     : 0.0.0.0
DNS         :
DHCP SERVER : 172.16.10.201
DHCP LEASE  : 0, 0/0/0
MAC         : 00:50:79:66:68:1e
LPORT       : 20000
RHOST:PORT  : 127.0.0.1:30000
MTU         : 1500

VPCS>
```

```
Neighbor ID     Pri   State           Dead Time   Address         Interface
172.16.11.4       0   FULL/  -        00:00:33    172.16.10.202   Ethernet0/1
172.16.11.3       0   FULL/  -        00:00:32    172.16.10.198   Ethernet0/0
172.16.11.2       0   FULL/  -        00:00:34    172.16.10.194   Serial1/0
R1-CORE#show ip int
R1-CORE#show ip interface br
R1-CORE#show ip interface brief
Interface                  IP-Address      OK? Method Status                Protocol
Ethernet0/0                172.16.10.197   YES manual up                    up
Ethernet0/1                172.16.10.201   YES manual up                    up
Ethernet0/2                203.0.113.2     YES manual up                    up
Ethernet0/3                unassigned      YES unset  administratively down down
Serial1/0                  172.16.10.193   YES manual up                    up
Serial1/1                  unassigned      YES unset  administratively down down
Serial1/2                  unassigned      YES unset  administratively down down
Serial1/3                  unassigned      YES unset  administratively down down
Serial2/0                  unassigned      YES unset  administratively down down
Serial2/1                  unassigned      YES unset  administratively down down
Serial2/2                  unassigned      YES unset  administratively down down
Serial2/3                  unassigned      YES unset  administratively down down
Loopback0                  172.16.11.1     YES manual up                    up
R1-CORE#
```

```
SWM1#show standby brief
                     P indicates configured to preempt.
                     |
Interface   Grp  Pri P State   Active          Standby         Virtual IP
Vl10        10   110 P Active  local           172.16.0.3      172.16.0.1
Vl20        20   100 P Standby 172.16.8.3      local           172.16.8.1
Vl30        30   110 P Active  local           172.16.10.67    172.16.10.65
Vl40        40   100 P Standby 172.16.10.131   local           172.16.10.129
SWM1#
```

```
SWM2#show standby brief
                     P indicates configured to preempt.
                     |
Interface   Grp  Pri P State   Active          Standby         Virtual IP
Vl10        10   100 P Standby 172.16.0.2      local           172.16.0.1
Vl20        20   110 P Active  local           172.16.8.2      172.16.8.1
Vl30        30   100 P Standby 172.16.10.66    local           172.16.10.65
Vl40        40   110 P Active  local           172.16.10.130   172.16.10.129
SWM2#
```

```
R1-CORE#ping 203.0.113.1
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 203.0.113.1, timeout is 2 seconds:
.!!!!
Success rate is 80 percent (4/5), round-trip min/avg/max = 1/1/1 ms
R1-CORE#
```

```
VPCS> ping 172.16.8.1

84 bytes from 172.16.8.1 icmp_seq=1 ttl=255 time=0.660 ms
84 bytes from 172.16.8.1 icmp_seq=2 ttl=255 time=0.794 ms
84 bytes from 172.16.8.1 icmp_seq=3 ttl=255 time=0.917 ms
84 bytes from 172.16.8.1 icmp_seq=4 ttl=255 time=0.838 ms
84 bytes from 172.16.8.1 icmp_seq=5 ttl=255 time=0.684 ms

VPCS> ping 172.16.10.10

172.16.10.10 icmp_seq=1 timeout
84 bytes from 172.16.10.10 icmp_seq=2 ttl=252 time=11.572 ms
84 bytes from 172.16.10.10 icmp_seq=3 ttl=252 time=10.823 ms
84 bytes from 172.16.10.10 icmp_seq=4 ttl=252 time=11.580 ms
84 bytes from 172.16.10.10 icmp_seq=5 ttl=252 time=10.229 ms
```

```
VPCS> ping 172.16.8.1

172.16.8.1 icmp_seq=1 timeout
84 bytes from 172.16.8.1 icmp_seq=2 ttl=255 time=1.576 ms
84 bytes from 172.16.8.1 icmp_seq=3 ttl=255 time=1.717 ms
84 bytes from 172.16.8.1 icmp_seq=4 ttl=255 time=1.369 ms
84 bytes from 172.16.8.1 icmp_seq=5 ttl=255 time=1.043 ms

VPCS>
```

```
VPCS> ping 172.16.10.10

84 bytes from 172.16.10.10 icmp_seq=1 ttl=254 time=1.800 ms
84 bytes from 172.16.10.10 icmp_seq=2 ttl=254 time=1.376 ms
84 bytes from 172.16.10.10 icmp_seq=3 ttl=254 time=1.387 ms
84 bytes from 172.16.10.10 icmp_seq=4 ttl=254 time=1.235 ms
84 bytes from 172.16.10.10 icmp_seq=5 ttl=254 time=1.527 ms

VPCS>
```

```
VPCS> ping 172.16.10.10

84 bytes from 172.16.10.10 icmp_seq=1 ttl=254 time=1.556 ms
84 bytes from 172.16.10.10 icmp_seq=2 ttl=254 time=1.424 ms
84 bytes from 172.16.10.10 icmp_seq=3 ttl=254 time=1.988 ms
84 bytes from 172.16.10.10 icmp_seq=4 ttl=254 time=1.664 ms
84 bytes from 172.16.10.10 icmp_seq=5 ttl=254 time=1.436 ms

VPCS>
```

```
VPCS> ping 172.16.9.1

84 bytes from 172.16.9.1 icmp_seq=1 ttl=253 time=10.982 ms
84 bytes from 172.16.9.1 icmp_seq=2 ttl=253 time=10.563 ms
84 bytes from 172.16.9.1 icmp_seq=3 ttl=253 time=7.171 ms
84 bytes from 172.16.9.1 icmp_seq=4 ttl=253 time=11.240 ms
84 bytes from 172.16.9.1 icmp_seq=5 ttl=253 time=10.673 ms

VPCS>
```

```
Enter configuration commands, one per line.  End with CNTL/Z.
R1-CORE(config)#ip ssh source-in
R1-CORE(config)#ip ssh source-interface loo
R1-CORE(config)#ip ssh source-interface loopback0
R1-CORE(config)#end
R1-CORE#ss
*Oct 25 05:04:48.695: %SYS-5-CONFIG_I: Configured from console by console
R1-CORE#ssh -l admin 172.16.11.3
Password:

SWM1#exit

[Connection to 172.16.11.3 closed by foreign host]
R1-CORE#no ip ssh sour
R1-CORE#no ip ssh source-in
R1-CORE#no ip ssh source-inter
R1-CORE#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R1-CORE(config)#no ip ssh sou
R1-CORE(config)#no ip ssh source-interface lo
R1-CORE(config)#no ip ssh source-interface loo
R1-CORE(config)#no ip ssh source-interface loopb
R1-CORE(config)#no ip ssh source-interface loopback 0
R1-CORE(config)#end
R1-CORE#ss
*Oct 25 05:06:05.837: %SYS-5-CONFIG_I: Configured from console by console
R1-CORE#ssh -l admin 172.16.11.3
% Connection refused by remote host

R1-CORE#
```

```
SWM1#show ip ssh
SSH Enabled - version 2.0
Authentication methods:publickey,keyboard-interactive,password
Authentication Publickey Algorithms:x509v3-ssh-rsa,ssh-rsa
Hostkey Algorithms:x509v3-ssh-rsa,ssh-rsa
Encryption Algorithms:aes128-ctr,aes192-ctr,aes256-ctr
MAC Algorithms:hmac-sha1,hmac-sha1-96
Authentication timeout: 120 secs; Authentication retries: 3
Minimum expected Diffie Hellman key size : 1024 bits
IOS Keys in SECSH format(ssh-rsa, base64 encoded): SWM1.itla.lab
ssh-rsa AAAAB3NzaC1yc2EAAAADAQABAAABAQC45ZRP6+u+rALsy8pUAQPQb5rh0Z1CKK1UKi1I3iyY
9VgpYVniaQxzo7J8Z+tRBZS+q4bWR8RW7HqSC4xq1dozglZMvzDBGkFSvV1FUGq4D/laoZZc3pII8NIG
4Q3XUEOCrJddeWLOV5cJdqQm2zjnCe72WGLK/HPAl6D57zMaOMra7976iLQxj35aJupfcc1rK7lAykMa
Bg8Uy7XUeGUsb1aHmBK/YBCsFDvKKkEfGzlpQneZIK65TwTEny4qqG+BlcSJ+F026Hm26bofK5hBso6B
B2PdwIlJ/v5v/o+HLrz2Srm9Gk0gTuGGPxEwDG887cW4UqcRF4cRHx4SjY1p
SWM1#show access-lists VTY-ADMIN
Standard IP access list VTY-ADMIN
    10 permit 172.16.10.64, wildcard bits 0.0.0.63
    20 permit 172.16.11.0, wildcard bits 0.0.0.255 (2 matches)
SWM1#
```

```
SWM2#show ip ssh
SSH Enabled - version 2.0
Authentication methods:publickey,keyboard-interactive,password
Authentication Publickey Algorithms:x509v3-ssh-rsa,ssh-rsa
Hostkey Algorithms:x509v3-ssh-rsa,ssh-rsa
Encryption Algorithms:aes128-ctr,aes192-ctr,aes256-ctr
MAC Algorithms:hmac-sha1,hmac-sha1-96
Authentication timeout: 120 secs; Authentication retries: 3
Minimum expected Diffie Hellman key size : 1024 bits
IOS Keys in SECSH format(ssh-rsa, base64 encoded): SWM2.itla.lab
ssh-rsa AAAAB3NzaC1yc2EAAAADAQABAAABAQDFyLvUXl6w7IQp83XgLnnRUjbJItNomLQ19lLGxvOA
PnMdjeA5WQVdM+9GHWQmbam7bCZ92/AT7735qZ90HhkB03PR2pQk4uUfNwW8m9K/16rJgvpVvEAZVU2p
ovCIWsTeIfKoPLYHQln50NKbbbN7+6gx0nGxuefyqRuQAtClDXbiR/d8oYZyTxKFybuQuw1K58c6nvtt
TBLQ/cL+sBRBIoWhGa7/9bZ/29eU86gtSoGuDpXKLrfbPUixpQL0MvCorvsDebYgNm4Bo2DuSwtgUDcw
DOdH5umb6jtjjYbV5HCbey0plwKDuD6nI2fk4yVyC1PHM3NQG5lndPpS+cT9
SWM2#show access-lists VTY-ADMIN
Standard IP access list VTY-ADMIN
    10 permit 172.16.10.64, wildcard bits 0.0.0.63
    20 permit 172.16.11.0, wildcard bits 0.0.0.255
SWM2#
```

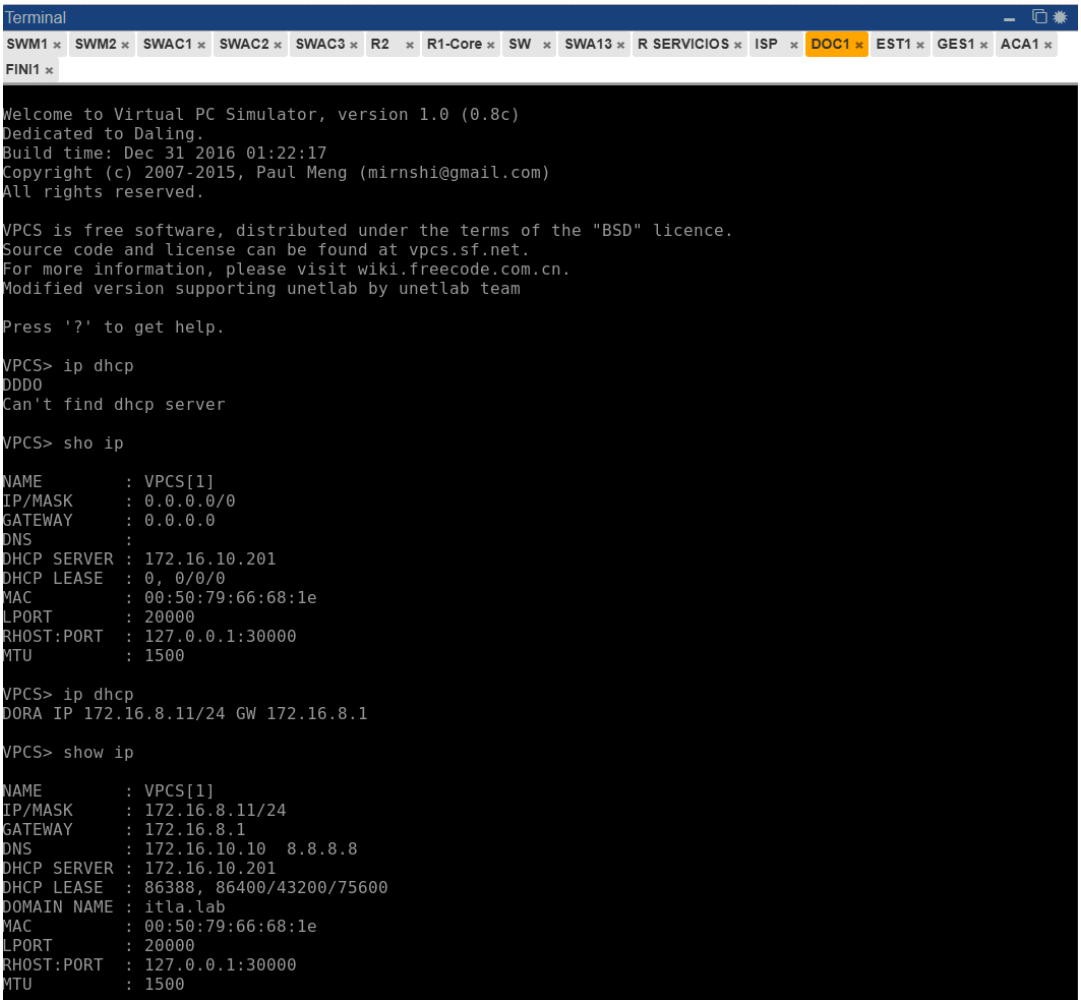
```
SWAC2#show ip ssh
SSH Enabled - version 2.0
Authentication methods:publickey,keyboard-interactive,password
Authentication Publickey Algorithms:x509v3-ssh-rsa,ssh-rsa
Hostkey Algorithms:x509v3-ssh-rsa,ssh-rsa
Encryption Algorithms:aes128-ctr,aes192-ctr,aes256-ctr
MAC Algorithms:hmac-sha1,hmac-sha1-96
Authentication timeout: 120 secs; Authentication retries: 3
Minimum expected Diffie Hellman key size : 1024 bits
IOS Keys in SECSH format(ssh-rsa, base64 encoded): SWAC2.itla.lab
ssh-rsa AAAAB3NzaC1yc2EAAAADAQABAAABAQCS2Ya+S2Fev+Uafvw1VwTvQJM56VCqFdyqVPoI7sQP
Cas+UtJaAv8jHndkIRtbnObsXWUc+rRibcsatLjH5wNyTpZ+hgHA2luy+D+pB0gOU00JYYItlYpVteMA
zAa6HtuVP9V+bRWYijZeBuS9IhTd3lJN2jM1YVEzw8hyntLChI2pdfnDSRgIAN27IFsHU5jXOecERwwN
UuJraWTbBzs4ysYUeB86e+TSTdj3Bror6LiMoQ6s7e7pbt8B0adupjRvAgDa0AB8hJGCmJTuAb6mwhKG
sorxRkzrqRZcram2tkzf2+VyE5CUXbV6uru/TrJBRksKRJhCq2tb6UVwShpD
SWAC2#show access-lists VTY-ADMIN
Standard IP access list VTY-ADMIN
    10 permit 172.16.10.64, wildcard bits 0.0.0.63
    20 permit 172.16.11.0, wildcard bits 0.0.0.255
SWAC2#
```

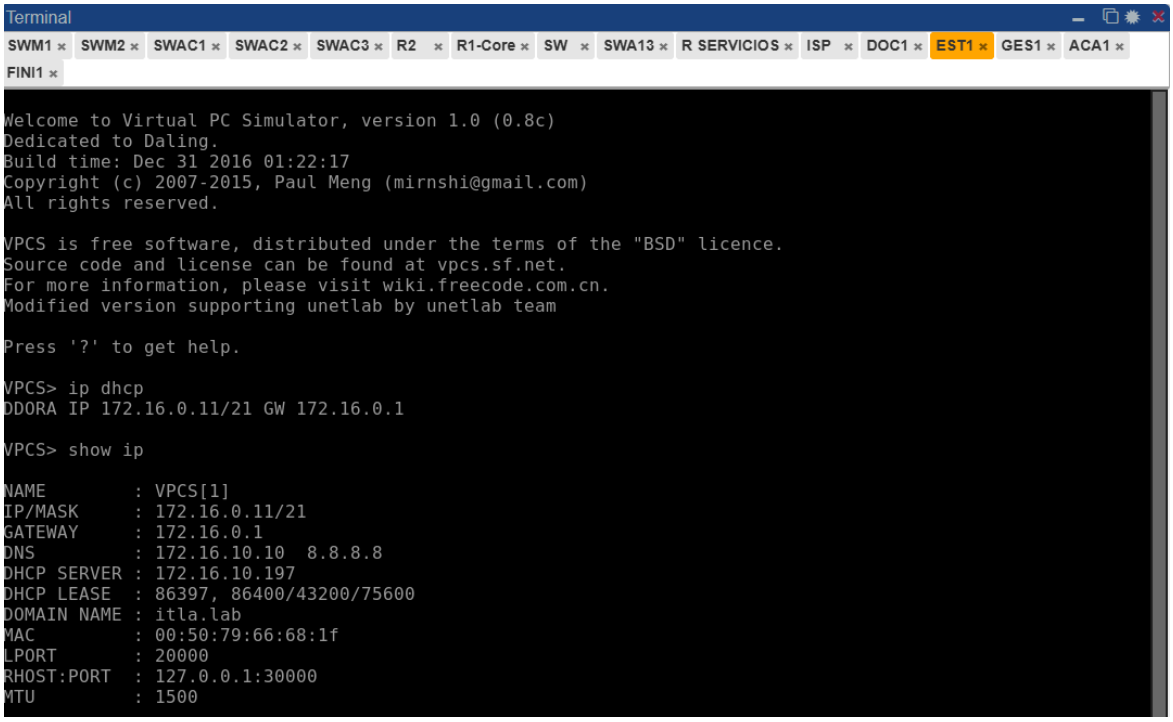
```
SWAC3#show ip ssh
SSH Enabled - version 2.0
Authentication methods:publickey,keyboard-interactive,password
Authentication Publickey Algorithms:x509v3-ssh-rsa,ssh-rsa
Hostkey Algorithms:x509v3-ssh-rsa,ssh-rsa
Encryption Algorithms:aes128-ctr,aes192-ctr,aes256-ctr
MAC Algorithms:hmac-sha1,hmac-sha1-96
Authentication timeout: 120 secs; Authentication retries: 3
Minimum expected Diffie Hellman key size : 1024 bits
IOS Keys in SECSH format(ssh-rsa, base64 encoded): TP-self-signed-67153925
ssh-rsa AAAAB3NzaC1yc2EAAAADAQABAAAAgQCpbyNV7+X5y1GSM/PZgODhkpCX4taMFgpevX0D67mX
6dmr3u4C6mj47qAj8vDj5FNCCnUlNCyaJDqPBe63h5LysIYnG7bKvRO+9fcbma3HjSebut1jNe+oORqP
m0s1pUk7GIPxptefZVqmjt35blo7+0srsGNND1AHXcZsDXutvw==
SWAC3#show access-lists VTY-ADMIN
Standard IP access list VTY-ADMIN
    10 permit 172.16.10.64, wildcard bits 0.0.0.63
    20 permit 172.16.11.0, wildcard bits 0.0.0.255
SWAC3#
```

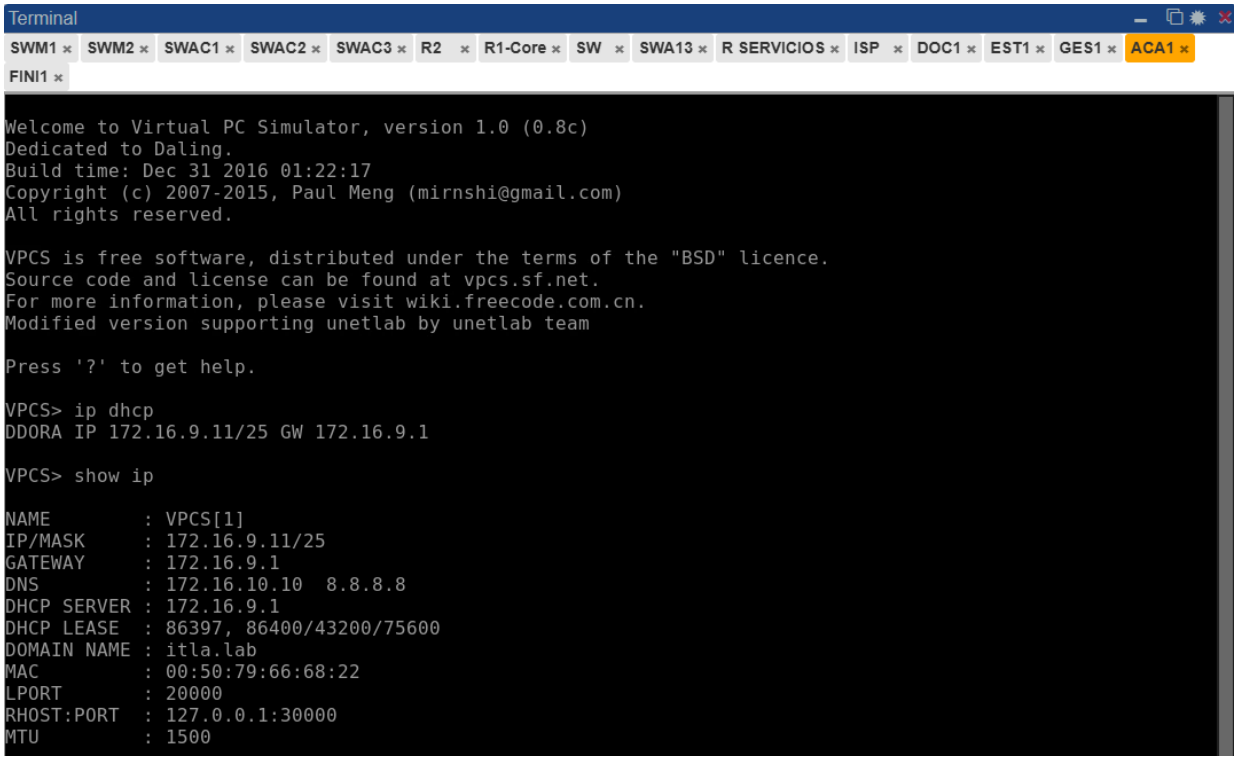
```
*Oct 25 05:05:29.712: %SYS-5-CONFIG_I: Configured from console by console
R2#show ip ssh
SSH Enabled - version 2.0
Authentication methods:publickey,keyboard-interactive,password
Authentication Publickey Algorithms:x509v3-ssh-rsa,ssh-rsa
Hostkey Algorithms:x509v3-ssh-rsa,ssh-rsa
Encryption Algorithms:aes128-ctr,aes192-ctr,aes256-ctr
MAC Algorithms:hmac-sha2-256,hmac-sha2-512,hmac-sha1,hmac-sha1-96
KEX Algorithms:diffie-hellman-group-exchange-sha1,diffie-hellman-group14-sha1
Authentication timeout: 120 secs; Authentication retries: 3
Minimum expected Diffie Hellman key size : 2048 bits
IOS Keys in SECSH format(ssh-rsa, base64 encoded): R2.itla.lab
ssh-rsa AAAAB3NzaC1yc2EAAAADAQABAAABAQCZMEhUvTFiLH69YvFrjmP39TP98USpQQTavxzidKQw
okvLcw3YOLrMP73HsJBe1qV2ziBXZ1gXeMJqIFQ4v1cILjZWPe6XEKBXiVanvNuop7wiOYvg11JQDMsa
UmP+wVIWFF96iDd3PLkLOE1pdiQQSHD+TbOtyhQCK1k3XoQHlrRwyIMrF8kQVnAmd/mmSpC1dWfZrAUY
egpT5rtNooxf+y7/D4+maeS8rFB6bf8dC/Bqi6NsbmEmfiEtBceEw8cFCZtfl6xy8bgNn5DziVQFj89l
EiAjveRTnQ1FjNHBMbJaIqNdpUxCVbwNPOSyjy7V8lP2Pk5T8PEiWs18xUZX
R2#show access-lists VTY-ADMIN
Standard IP access list VTY-ADMIN
    10 permit 172.16.10.64, wildcard bits 0.0.0.63
    20 permit 172.16.11.0, wildcard bits 0.0.0.255
R2#
```

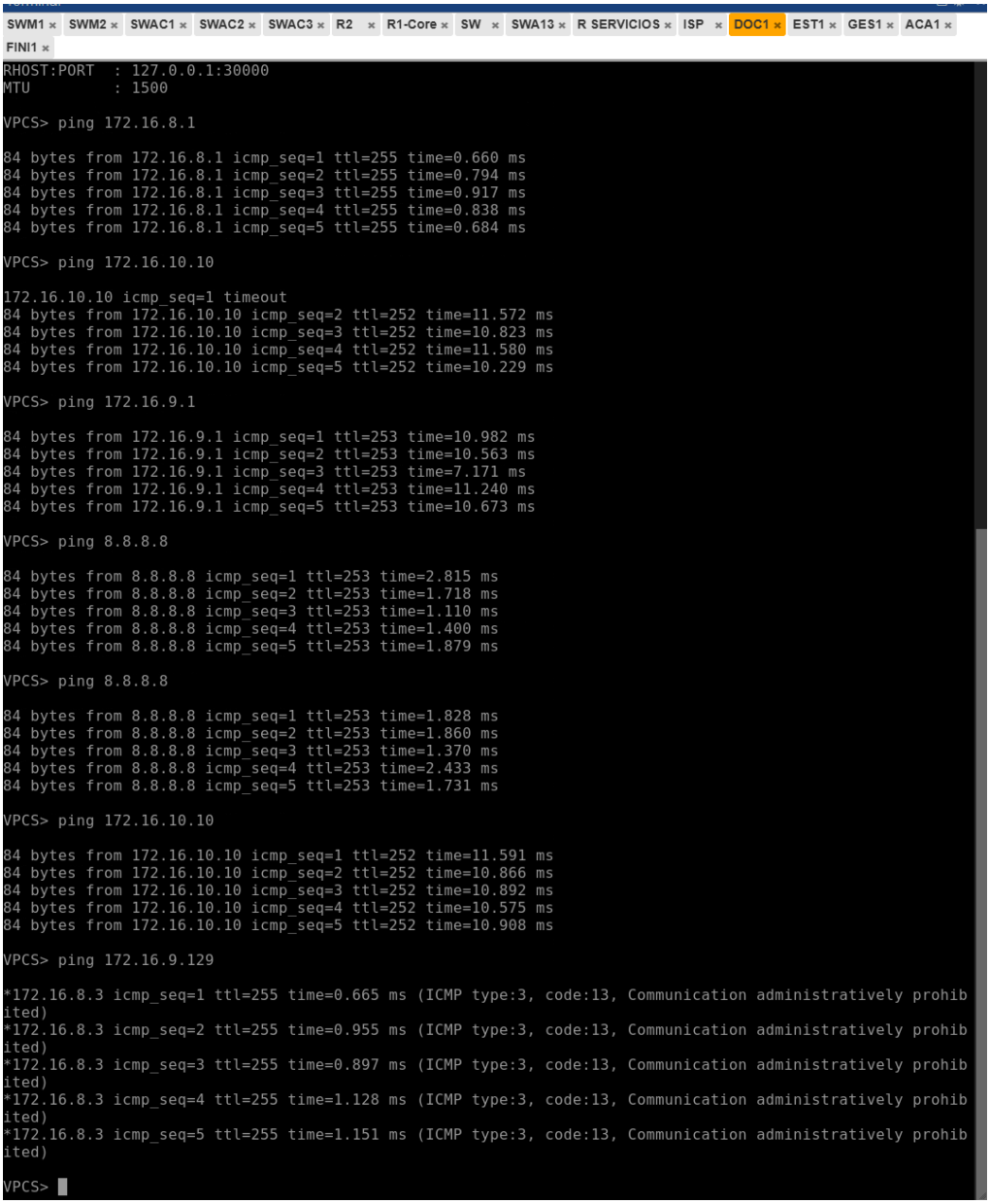
```
R1-CORE#show ip ssh
SSH Enabled - version 2.0
Authentication methods:publickey,keyboard-interactive,password
Authentication Publickey Algorithms:x509v3-ssh-rsa,ssh-rsa
Hostkey Algorithms:x509v3-ssh-rsa,ssh-rsa
Encryption Algorithms:aes128-ctr,aes192-ctr,aes256-ctr
MAC Algorithms:hmac-sha2-256,hmac-sha2-512,hmac-sha1,hmac-sha1-96
KEX Algorithms:diffie-hellman-group-exchange-sha1,diffie-hellman-group14-sha1
Authentication timeout: 120 secs; Authentication retries: 3
Minimum expected Diffie Hellman key size : 2048 bits
IOS Keys in SECSH format(ssh-rsa, base64 encoded): R1-CORE.itla.lab
ssh-rsa AAAAB3NzaC1yc2EAAAADAQABAAABAQCmJEtxWW+ovCeJ5ZXA+WeNNobUPdIIQ/SSGvF8LHha
q9zx/bxNraCABMyV7Cd9ekG/6Y6dFtsRaktvz81nS1fKMR8XffzTfgHRo20crCUr0AbpDIu1fIrc3lNr
bv2I5/jvUJZfAPUuYI+TLWPkUXLPe5Q3Aw4D+zmETHX3/Cvyr51SU2vLkY6+XtbatDVh/ZxHlo1UNNt5
lcLvGAx3WN+N3IDfyNettV1urMDjYWLPhb+JRbLwiklqOqj60j0A4kcCNB9JywHEplcPcH4F4ru++fc8
BweNx1cFpWDGh1571OFHR5mPzTwNW9XnL3JzBAJVd/T3InUYoBx0bdkT4O4J
R1-CORE#show access-lists VTY-ADMIN
Standard IP access list VTY-ADMIN
    10 permit 172.16.10.64, wildcard bits 0.0.0.63
    20 permit 172.16.11.0, wildcard bits 0.0.0.255
R1-CORE#
```

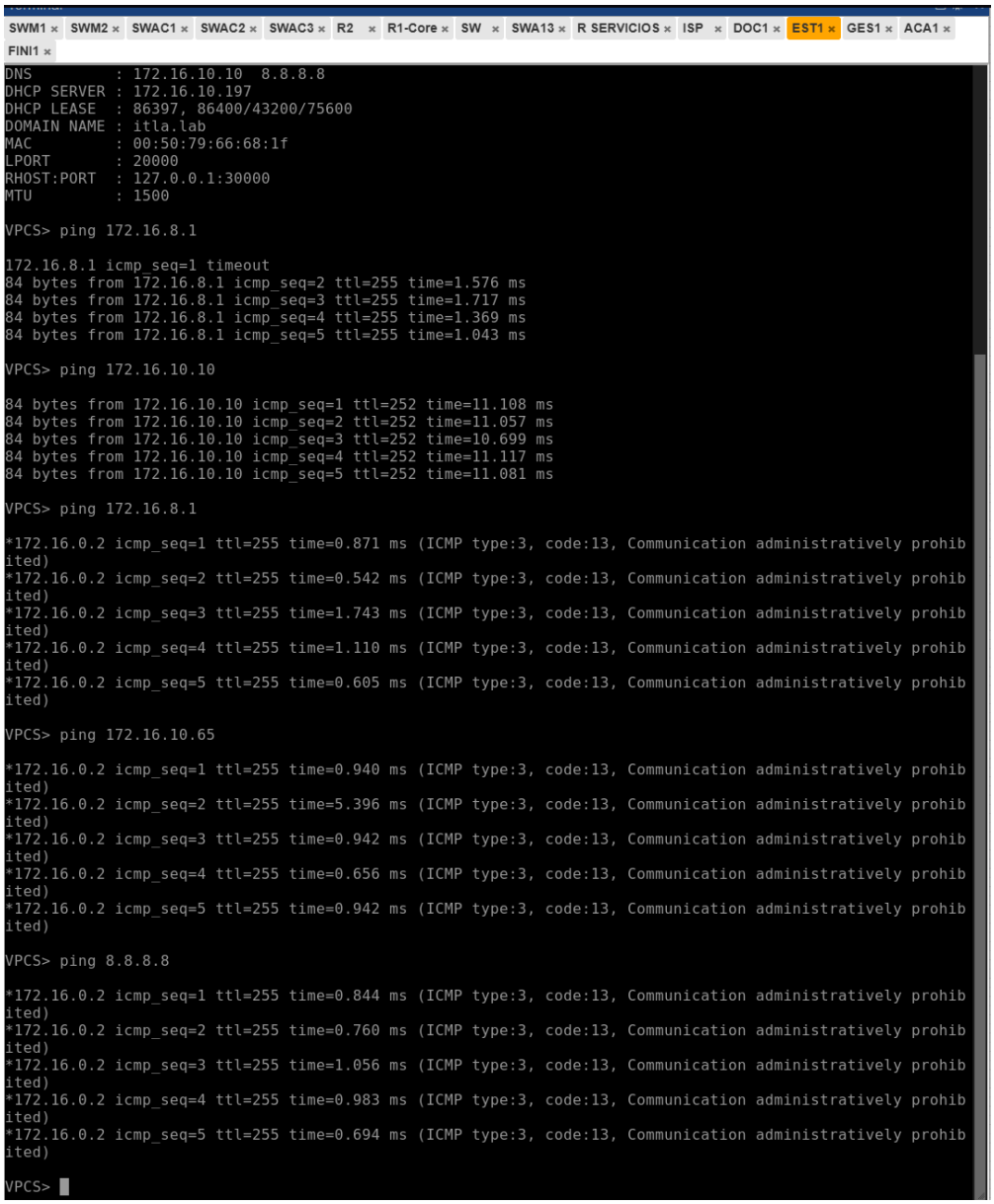
```
, with SERVICIO Ethernet0/0 (half duplex).
SW#show ip ssh
SSH Enabled - version 2.0
Authentication methods:publickey,keyboard-interactive,password
Authentication Publickey Algorithms:x509v3-ssh-rsa,ssh-rsa
Hostkey Algorithms:x509v3-ssh-rsa,ssh-rsa
Encryption Algorithms:aes128-ctr,aes192-ctr,aes256-ctr
MAC Algorithms:hmac-sha1,hmac-sha1-96
Authentication timeout: 120 secs; Authentication retries: 3
Minimum expected Diffie Hellman key size : 1024 bits
IOS Keys in SECSH format(ssh-rsa, base64 encoded): SW.itla.lab
ssh-rsa AAAAB3NzaC1yc2EAAAADAQABAAABAQCxDhHYuPxsRAXuwY2CsgQUduhXyKObYT6MxhFjQa37
uKXAreaiIkZNbvYbtCuHX3KjckO2+WskoapLWa4V73UWs8hVck2u6DeE4G8xwLJ6OYdrtCJNs4pPIENj
NJizb1lYQccca2TBmkITGPbApmZmt7tfV4mhYoyLAWclt6MzcWUYxaZbeMPUOaQF4Gi12Y594qvguUWK
SQmsLGNSMqFw/KjeC626SFcHQ9iEGYXqh1I86E0pG3aLOXuJL2/JbPqDkdC+Okj9DHs1b5zVsTlc4QQr
gEqfPIaWKSZiBZbf5Z0Qw0R4ZRMT0OHO7hW+oEogjuR3bwxO7wlyQckLnBn1
SW#show access-lists VTY-ADMIN
Standard IP access list VTY-ADMIN
    10 permit 172.16.10.64, wildcard bits 0.0.0.63
    20 permit 172.16.11.0, wildcard bits 0.0.0.255
SW#
```

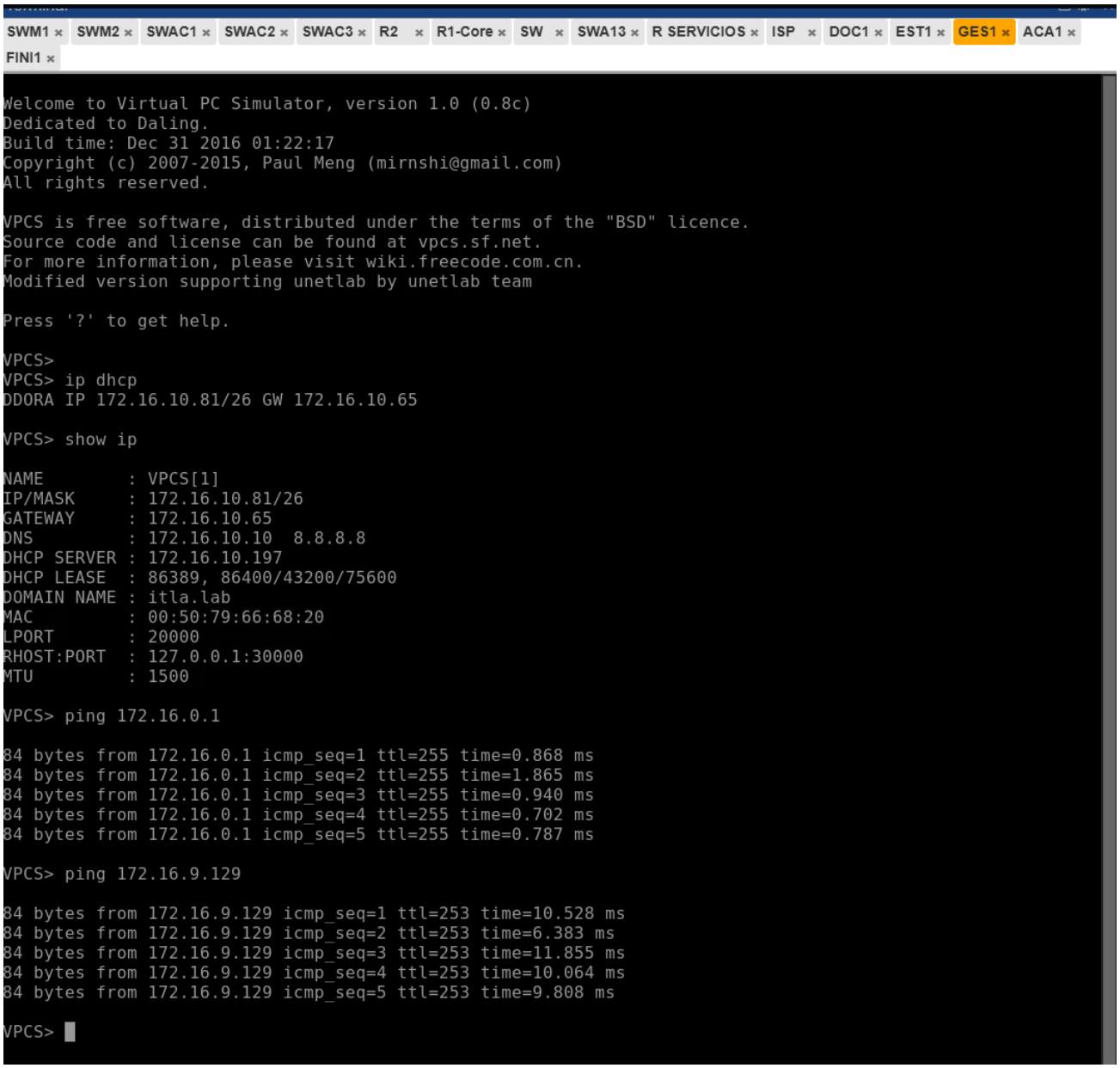

Terminal
SWM1 × SWM2 × SWAC1 × SWAC2 × SWAC3 × R2 × R1-Core × SW × SWA13 × R SERVICIOS × ISP × DOC1 × EST1 × GES1 × ACA1 ×
FINI1 ×
Welcome to Virtual PC Simulator, version 1.0 (0.8c)
Dedicated to Daling.
Build time: Dec 31 2016 01:22:17
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Source code and license can be found at vpcs.sf.net.
For more information, please visit wiki.freecode.com.cn.
Modified version supporting unetlab by unetlab team
Press '?' to get help.
VPCS> ip dhcp
DDDO
Can't find dhcp server
VPCS> sho ip
NAME : VPCS[1]
IP/MASK : 0.0.0.0/0
GATEWAY : 0.0.0.0
DNS :
DHCP SERVER : 172.16.10.201
DHCP LEASE : 0, 0/0/0
MAC : 00:50:79:66:68:1e
LPORT : 20000
RHOST:PORT : 127.0.0.1:30000
MTU : 1500
VPCS> ip dhcp
DORA IP 172.16.8.11/24 GW 172.16.8.1
VPCS> show ip
NAME : VPCS[1]
IP/MASK : 172.16.8.11/24
GATEWAY : 172.16.8.1
DNS : 172.16.10.10 8.8.8.8
DHCP SERVER : 172.16.10.201
DHCP LEASE : 86388, 86400/43200/75600
DOMAIN NAME : itla.lab
MAC : 00:50:79:66:68:1e
LPORT : 20000
RHOST:PORT : 127.0.0.1:30000
MTU : 1500

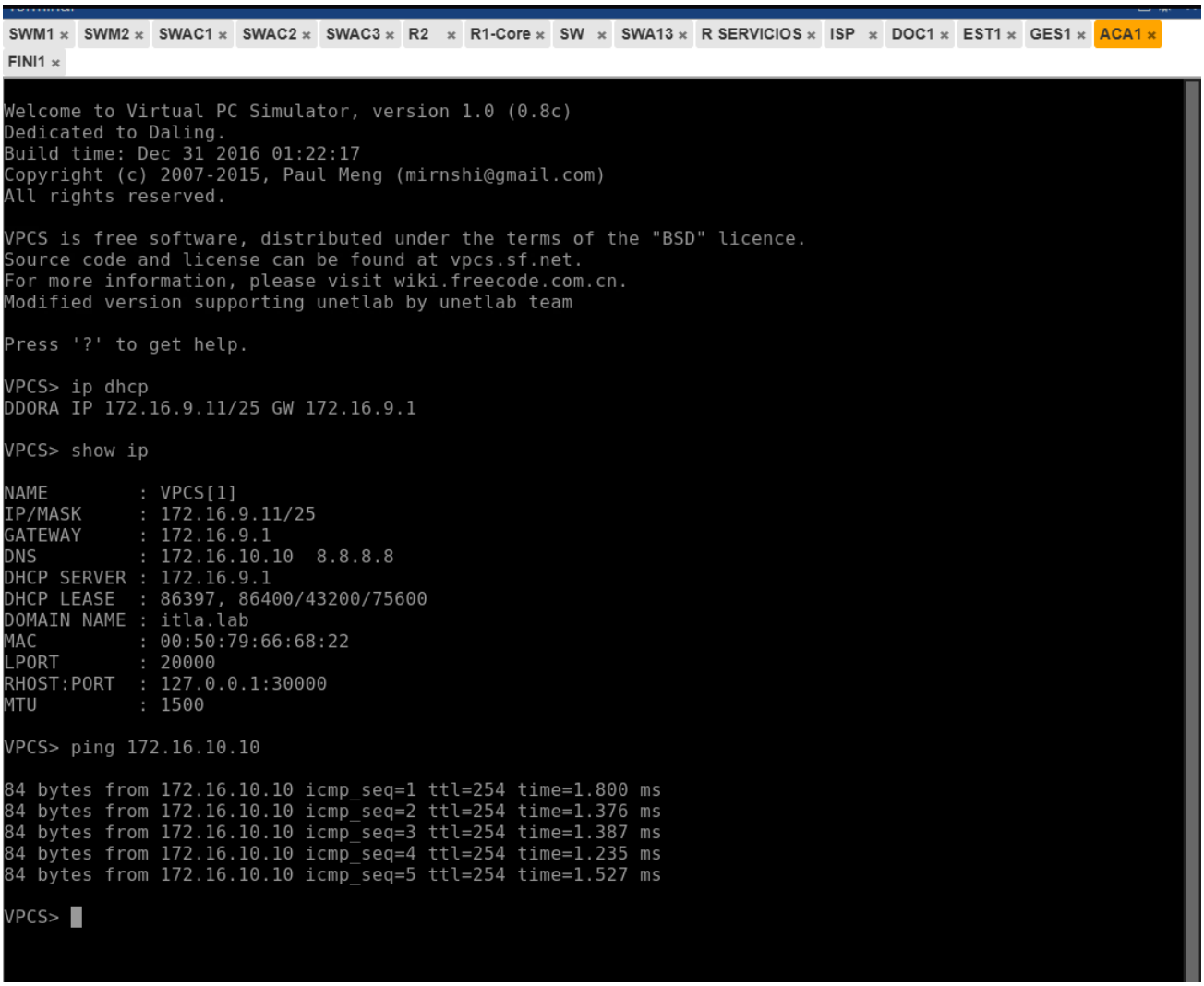

Terminal
SWM1 × SWM2 × SWAC1 × SWAC2 × SWAC3 × R2 × R1-Core × SW × SWA13 × R SERVICIOS × ISP × DOC1 × EST1 × GES1 × ACA1 ×
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For more information, please visit wiki.freecode.com.cn.
Modified version supporting unetlab by unetlab team
Press '?' to get help.
VPCS> ip dhcp
DDORA IP 172.16.0.11/21 GW 172.16.0.1
VPCS> show ip
NAME : VPCS[1]
IP/MASK : 172.16.0.11/21
GATEWAY : 172.16.0.1
DNS : 172.16.10.10 8.8.8.8
DHCP SERVER : 172.16.10.197
DHCP LEASE : 86397, 86400/43200/75600
DOMAIN NAME : itla.lab
MAC : 00:50:79:66:68:1f
LPORT : 20000
RHOST:PORT : 127.0.0.1:30000
MTU : 1500


Terminal
SWM1 × SWM2 × SWAC1 × SWAC2 × SWAC3 × R2 × R1-Core × SW × SWA13 × R SERVICIOS × ISP × DOC1 × EST1 × GES1 × ACA1 ×
FINI1 ×
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For more information, please visit wiki.freecode.com.cn.
Modified version supporting unetlab by unetlab team
Press '?' to get help.
VPCS> ip dhcp
DDORA IP 172.16.9.11/25 GW 172.16.9.1
VPCS> show ip
NAME : VPCS[1]
IP/MASK : 172.16.9.11/25
GATEWAY : 172.16.9.1
DNS : 172.16.10.10 8.8.8.8
DHCP SERVER : 172.16.9.1
DHCP LEASE : 86397, 86400/43200/75600
DOMAIN NAME : itla.lab
MAC : 00:50:79:66:68:22
LPORT : 20000
RHOST:PORT : 127.0.0.1:30000
MTU : 1500


SWM1 × SWM2 × SWAC1 × SWAC2 × SWAC3 × R2 × R1-Core × SW × SWA13 × R SERVICIOS × ISP × DOC1 × EST1 × GES1 × ACA1 × FINI1 ×
RHOST:PORT : 127.0.0.1:30000
MTU : 1500
VPCS> ping 172.16.8.1
84 bytes from 172.16.8.1 icmp_seq=1 ttl=255 time=0.660 ms
84 bytes from 172.16.8.1 icmp_seq=2 ttl=255 time=0.794 ms
84 bytes from 172.16.8.1 icmp_seq=3 ttl=255 time=0.917 ms
84 bytes from 172.16.8.1 icmp_seq=4 ttl=255 time=0.838 ms
84 bytes from 172.16.8.1 icmp_seq=5 ttl=255 time=0.684 ms
VPCS> ping 172.16.10.10
172.16.10.10 icmp_seq=1 timeout
84 bytes from 172.16.10.10 icmp_seq=2 ttl=252 time=11.572 ms
84 bytes from 172.16.10.10 icmp_seq=3 ttl=252 time=10.823 ms
84 bytes from 172.16.10.10 icmp_seq=4 ttl=252 time=11.580 ms
84 bytes from 172.16.10.10 icmp_seq=5 ttl=252 time=10.229 ms
VPCS> ping 172.16.9.1
84 bytes from 172.16.9.1 icmp_seq=1 ttl=253 time=10.982 ms
84 bytes from 172.16.9.1 icmp_seq=2 ttl=253 time=10.563 ms
84 bytes from 172.16.9.1 icmp_seq=3 ttl=253 time=7.171 ms
84 bytes from 172.16.9.1 icmp_seq=4 ttl=253 time=11.240 ms
84 bytes from 172.16.9.1 icmp_seq=5 ttl=253 time=10.673 ms
VPCS> ping 8.8.8.8
84 bytes from 8.8.8.8 icmp_seq=1 ttl=253 time=2.815 ms
84 bytes from 8.8.8.8 icmp_seq=2 ttl=253 time=1.718 ms
84 bytes from 8.8.8.8 icmp_seq=3 ttl=253 time=1.110 ms
84 bytes from 8.8.8.8 icmp_seq=4 ttl=253 time=1.400 ms
84 bytes from 8.8.8.8 icmp_seq=5 ttl=253 time=1.879 ms
VPCS> ping 8.8.8.8
84 bytes from 8.8.8.8 icmp_seq=1 ttl=253 time=1.828 ms
84 bytes from 8.8.8.8 icmp_seq=2 ttl=253 time=1.860 ms
84 bytes from 8.8.8.8 icmp_seq=3 ttl=253 time=1.370 ms
84 bytes from 8.8.8.8 icmp_seq=4 ttl=253 time=2.433 ms
84 bytes from 8.8.8.8 icmp_seq=5 ttl=253 time=1.731 ms
VPCS> ping 172.16.10.10
84 bytes from 172.16.10.10 icmp_seq=1 ttl=252 time=11.591 ms
84 bytes from 172.16.10.10 icmp_seq=2 ttl=252 time=10.866 ms
84 bytes from 172.16.10.10 icmp_seq=3 ttl=252 time=10.892 ms
84 bytes from 172.16.10.10 icmp_seq=4 ttl=252 time=10.575 ms
84 bytes from 172.16.10.10 icmp_seq=5 ttl=252 time=10.908 ms
VPCS> ping 172.16.9.129
*172.16.8.3 icmp_seq=1 ttl=255 time=0.665 ms (ICMP type:3, code:13, Communication administratively prohibited)
*172.16.8.3 icmp_seq=2 ttl=255 time=0.955 ms (ICMP type:3, code:13, Communication administratively prohibited)
*172.16.8.3 icmp_seq=3 ttl=255 time=0.897 ms (ICMP type:3, code:13, Communication administratively prohibited)
*172.16.8.3 icmp_seq=4 ttl=255 time=1.128 ms (ICMP type:3, code:13, Communication administratively prohibited)
*172.16.8.3 icmp_seq=5 ttl=255 time=1.151 ms (ICMP type:3, code:13, Communication administratively prohibited)
VPCS>


SWM1 × SWM2 × SWAC1 × SWAC2 × SWAC3 × R2 × R1-Core × SW × SWA13 × R SERVICIOS × ISP × DOC1 × EST1 × GES1 × ACA1 × FINI1 ×
DNS : 172.16.10.10 8.8.8.8
DHCP SERVER : 172.16.10.197
DHCP LEASE : 86397, 86400/43200/75600
DOMAIN NAME : itla.lab
MAC : 00:50:79:66:68:1f
LPORT : 20000
RHOST:PORT : 127.0.0.1:30000
MTU : 1500
VPCS> ping 172.16.8.1
172.16.8.1 icmp_seq=1 timeout
84 bytes from 172.16.8.1 icmp_seq=2 ttl=255 time=1.576 ms
84 bytes from 172.16.8.1 icmp_seq=3 ttl=255 time=1.717 ms
84 bytes from 172.16.8.1 icmp_seq=4 ttl=255 time=1.369 ms
84 bytes from 172.16.8.1 icmp_seq=5 ttl=255 time=1.043 ms
VPCS> ping 172.16.10.10
84 bytes from 172.16.10.10 icmp_seq=1 ttl=252 time=11.108 ms
84 bytes from 172.16.10.10 icmp_seq=2 ttl=252 time=11.057 ms
84 bytes from 172.16.10.10 icmp_seq=3 ttl=252 time=10.699 ms
84 bytes from 172.16.10.10 icmp_seq=4 ttl=252 time=11.117 ms
84 bytes from 172.16.10.10 icmp_seq=5 ttl=252 time=11.081 ms
VPCS> ping 172.16.8.1
*172.16.0.2 icmp_seq=1 ttl=255 time=0.871 ms (ICMP type:3, code:13, Communication administratively prohibited)
*172.16.0.2 icmp_seq=2 ttl=255 time=0.542 ms (ICMP type:3, code:13, Communication administratively prohibited)
*172.16.0.2 icmp_seq=3 ttl=255 time=1.743 ms (ICMP type:3, code:13, Communication administratively prohibited)
*172.16.0.2 icmp_seq=4 ttl=255 time=1.110 ms (ICMP type:3, code:13, Communication administratively prohibited)
*172.16.0.2 icmp_seq=5 ttl=255 time=0.605 ms (ICMP type:3, code:13, Communication administratively prohibited)
VPCS> ping 172.16.10.65
*172.16.0.2 icmp_seq=1 ttl=255 time=0.940 ms (ICMP type:3, code:13, Communication administratively prohibited)
*172.16.0.2 icmp_seq=2 ttl=255 time=5.396 ms (ICMP type:3, code:13, Communication administratively prohibited)
*172.16.0.2 icmp_seq=3 ttl=255 time=0.942 ms (ICMP type:3, code:13, Communication administratively prohibited)
*172.16.0.2 icmp_seq=4 ttl=255 time=0.656 ms (ICMP type:3, code:13, Communication administratively prohibited)
*172.16.0.2 icmp_seq=5 ttl=255 time=0.942 ms (ICMP type:3, code:13, Communication administratively prohibited)
VPCS> ping 8.8.8.8
*172.16.0.2 icmp_seq=1 ttl=255 time=0.844 ms (ICMP type:3, code:13, Communication administratively prohibited)
*172.16.0.2 icmp_seq=2 ttl=255 time=0.760 ms (ICMP type:3, code:13, Communication administratively prohibited)
*172.16.0.2 icmp_seq=3 ttl=255 time=1.056 ms (ICMP type:3, code:13, Communication administratively prohibited)
*172.16.0.2 icmp_seq=4 ttl=255 time=0.983 ms (ICMP type:3, code:13, Communication administratively prohibited)
*172.16.0.2 icmp_seq=5 ttl=255 time=0.694 ms (ICMP type:3, code:13, Communication administratively prohibited)
VPCS>


SWM1 × SWM2 × SWAC1 × SWAC2 × SWAC3 × R2 × R1-Core × SW × SWA13 × R SERVICIOS × ISP × DOC1 × EST1 × GES1 × ACA1 × FINI1 ×
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For more information, please visit wiki.freecode.com.cn.
Modified version supporting unetlab by unetlab team
Press '?' to get help.
VPCS>
VPCS> ip dhcp
DDORA IP 172.16.10.81/26 GW 172.16.10.65
VPCS> show ip
NAME : VPCS[1]
IP/MASK : 172.16.10.81/26
GATEWAY : 172.16.10.65
DNS : 172.16.10.10 8.8.8.8
DHCP SERVER : 172.16.10.197
DHCP LEASE : 86389, 86400/43200/75600
DOMAIN NAME : itla.lab
MAC : 00:50:79:66:68:20
LPORT : 20000
RHOST:PORT : 127.0.0.1:30000
MTU : 1500
VPCS> ping 172.16.0.1
84 bytes from 172.16.0.1 icmp_seq=1 ttl=255 time=0.868 ms
84 bytes from 172.16.0.1 icmp_seq=2 ttl=255 time=1.865 ms
84 bytes from 172.16.0.1 icmp_seq=3 ttl=255 time=0.940 ms
84 bytes from 172.16.0.1 icmp_seq=4 ttl=255 time=0.702 ms
84 bytes from 172.16.0.1 icmp_seq=5 ttl=255 time=0.787 ms
VPCS> ping 172.16.9.129
84 bytes from 172.16.9.129 icmp_seq=1 ttl=253 time=10.528 ms
84 bytes from 172.16.9.129 icmp_seq=2 ttl=253 time=6.383 ms
84 bytes from 172.16.9.129 icmp_seq=3 ttl=253 time=11.855 ms
84 bytes from 172.16.9.129 icmp_seq=4 ttl=253 time=10.064 ms
84 bytes from 172.16.9.129 icmp_seq=5 ttl=253 time=9.808 ms
VPCS>


SWM1 × SWM2 × SWAC1 × SWAC2 × SWAC3 × R2 × R1-Core × SW × SWA13 × R SERVICIOS × ISP × DOC1 × EST1 × GES1 × ACA1 ×
FINI1 ×
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For more information, please visit wiki.freecode.com.cn.
Modified version supporting unetlab by unetlab team
Press '?' to get help.
VPCS> ip dhcp
DDORA IP 172.16.9.11/25 GW 172.16.9.1
VPCS> show ip
NAME : VPCS[1]
IP/MASK : 172.16.9.11/25
GATEWAY : 172.16.9.1
DNS : 172.16.10.10 8.8.8.8
DHCP SERVER : 172.16.9.1
DHCP LEASE : 86397, 86400/43200/75600
DOMAIN NAME : itla.lab
MAC : 00:50:79:66:68:22
LPORT : 20000
RHOST:PORT : 127.0.0.1:30000
MTU : 1500
VPCS> ping 172.16.10.10
84 bytes from 172.16.10.10 icmp_seq=1 ttl=254 time=1.800 ms
84 bytes from 172.16.10.10 icmp_seq=2 ttl=254 time=1.376 ms
84 bytes from 172.16.10.10 icmp_seq=3 ttl=254 time=1.387 ms
84 bytes from 172.16.10.10 icmp_seq=4 ttl=254 time=1.235 ms
84 bytes from 172.16.10.10 icmp_seq=5 ttl=254 time=1.527 ms
VPCS>

```
R2(config)#router ospf 1
R2(config-router)# no network 172.16.9.0 0.0.0.127 area 0
R2(config-router)# no network 172.16.9.128 0.0.0.127 area 0
R2(config-router)# no network 172.16.10.0 0.0.0.63 area 0
R2(config-router)# no network 172.16.10.192 0.0.0.3 area 0
R2(config-router)# no network 172.16.11.2 0.0.0.0 area 0
R2(config-router)# network 172.16.9.0 0.0.0.127 area 1
R2(config-router)# network 172.16.9.128 0.0.0.127 area 1
R2(config-router)# network 172.16.10.0 0.0.0.63 area 1
R2(config-router)# network 172.16.10.192 0.0.0.3 area 1
R2(config-router)# network 172.16.11.2 0.0.0.0 area 1
R2(config-router)# area 1 stub
R2(config-router)# exit
R2(config)#end
*Oct 25 05:35:27.520: %OSPF-5-ADJCHG: Process 1, Nbr 172.16.11.1 on Serial1/0 from FULL
 to DOWN, Neighbor Down: Interface down or detached
R2(config)#end
*Oct 25 05:35:27.845: %OSPF-4-ERRRCV: Received invalid packet: mismatched area ID from
backbone area from 172.16.10.193, Serial1/0
R2(config)#end
R2#
*Oct 25 05:35:29.239: %SYS-5-CONFIG_I: Configured from console by console
R2#
*Oct 25 05:35:37.487: %OSPF-4-ERRRCV: Received invalid packet: mismatched area ID from
backbone area from 172.16.10.193, Serial1/0
R2#
*Oct 25 05:35:46.728: %OSPF-4-ERRRCV: Received invalid packet: mismatched area ID from
backbone area from 172.16.10.193, Serial1/0
R2#
*Oct 25 05:35:56.075: %OSPF-4-ERRRCV: Received invalid packet: mismatched area ID from
backbone area from 172.16.10.193, Serial1/0
R2#
*Oct 25 05:36:05.502: %OSPF-4-ERRRCV: Received invalid packet: mismatched area ID from
backbone area from 172.16.10.193, Serial1/0
R2#
*Oct 25 05:36:15.254: %OSPF-4-ERRRCV: Received invalid packet: mismatched area ID from
backbone area from 172.16.10.193, Serial1/0
R2#
*Oct 25 05:36:24.728: %OSPF-4-ERRRCV: Received invalid packet: mismatched area ID from
backbone area from 172.16.10.193, Serial1/0
R2#
*Oct 25 05:36:34.170: %OSPF-4-ERRRCV: Received invalid packet: mismatched area ID from
backbone area from 172.16.10.193, Serial1/0
R2#
*Oct 25 05:36:44.059: %OSPF-4-ERRRCV: Received invalid packet: mismatched area ID from
backbone area from 172.16.10.193, Serial1/0
R2#
```

```
R1-CORE#show ip ospf ne
R1-CORE#show ip ospf neighbor

Neighbor ID     Pri   State           Dead Time   Address         Interface
172.16.11.4       0   FULL/  -        00:00:34    172.16.10.202   Ethernet0/1
172.16.11.3       0   FULL/  -        00:00:35    172.16.10.198   Ethernet0/0
R1-CORE#
```

```
R1-CORE#show ip ospf interface serial1/0
Serial1/0 is up, line protocol is up
  Internet Address 172.16.10.193/30, Area 0, Attached via Network Statement
  Process ID 1, Router ID 172.16.11.1, Network Type POINT_TO_POINT, Cost: 647
  Topology-MTID    Cost    Disabled    Shutdown      Topology Name
        0           647       no          no            Base
  Transmit Delay is 1 sec, State POINT_TO_POINT
  Timer intervals configured, Hello 10, Dead 40, Wait 40, Retransmit 5
    oob-resync timeout 40
    Hello due in 00:00:04
  Supports Link-local Signaling (LLS)
  Cisco NSF helper support enabled
  IETF NSF helper support enabled
  Index 1/1/1, flood queue length 0
  Next 0x0(0)/0x0(0)/0x0(0)
  Last flood scan length is 1, maximum is 2
  Last flood scan time is 0 msec, maximum is 0 msec
  Neighbor Count is 0, Adjacent neighbor count is 0
  Suppress hello for 0 neighbor(s)
  Cryptographic authentication enabled
    Youngest key id is 1
R1-CORE#
```

```
R1-CORE#show ip ospf interface serial1/0
Serial1/0 is up, line protocol is up
  Internet Address 172.16.10.193/30, Area 0, Attached via Network Statement
  Process ID 1, Router ID 172.16.11.1, Network Type POINT_TO_POINT, Cost: 647
  Topology-MTID    Cost    Disabled    Shutdown      Topology Name
        0           647       no          no            Base
  Transmit Delay is 1 sec, State POINT_TO_POINT
  Timer intervals configured, Hello 10, Dead 40, Wait 40, Retransmit 5
    oob-resync timeout 40
    Hello due in 00:00:04
  Supports Link-local Signaling (LLS)
  Cisco NSF helper support enabled
  IETF NSF helper support enabled
  Index 1/1/1, flood queue length 0
  Next 0x0(0)/0x0(0)/0x0(0)
  Last flood scan length is 1, maximum is 2
  Last flood scan time is 0 msec, maximum is 0 msec
  Neighbor Count is 0, Adjacent neighbor count is 0
  Suppress hello for 0 neighbor(s)
  Cryptographic authentication enabled
    Youngest key id is 1
R1-CORE#show ip ospf neighbor

Neighbor ID     Pri   State           Dead Time   Address         Interface
172.16.11.4       0   FULL/  -        00:00:33    172.16.10.202   Ethernet0/1
172.16.11.3       0   FULL/  -        00:00:31    172.16.10.198   Ethernet0/0
R1-CORE#
```

En esta parte se van a adjuntar los links:

Link de video:

https://drive.google.com/file/d/1lS3Azz5vcjwtnh5ZpnZakpvwAtYZLbeW/view?usp=sharing